Uncovering simultaneous scientific breakthroughs with a robust measure of disruptiveness

Munjung Kim, Sadamori Kojaku, Yong-Yeol Ahn. Science Advances.



Munjung Kim



Sadamori Kojaku

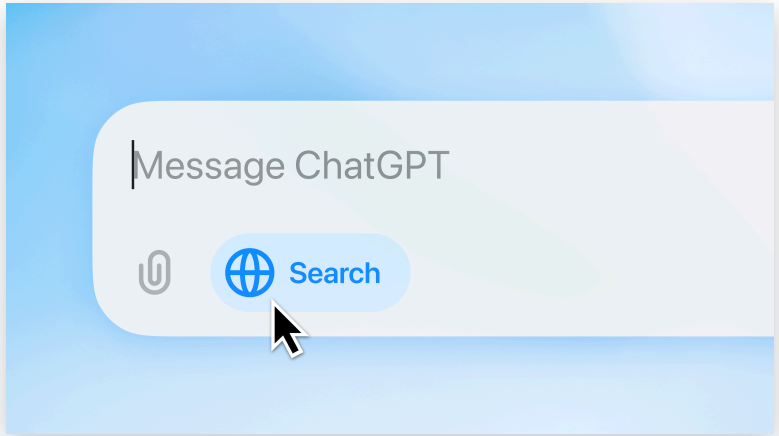


Yong-Yeol Ahn

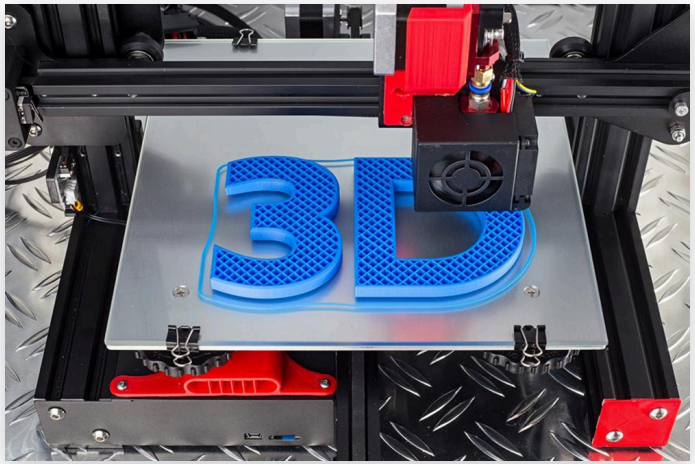
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BINGHAMTON UNIVERSITY
STATE UNIVERSITY OF NEW YORK

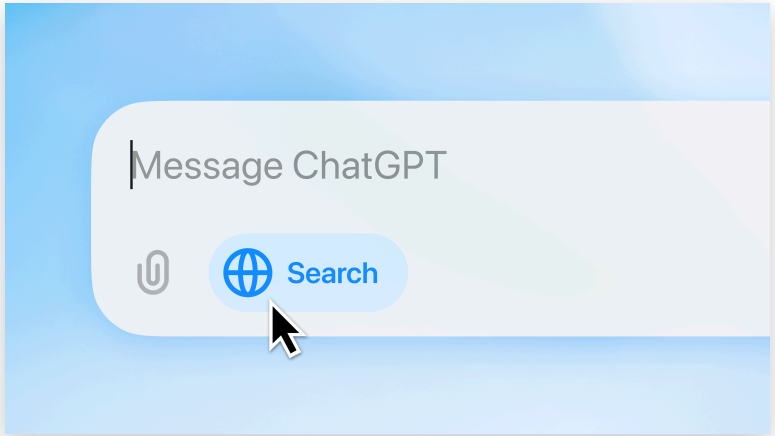

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Google
Duo
Assistant
Play Store


Message ChatGPT
Search

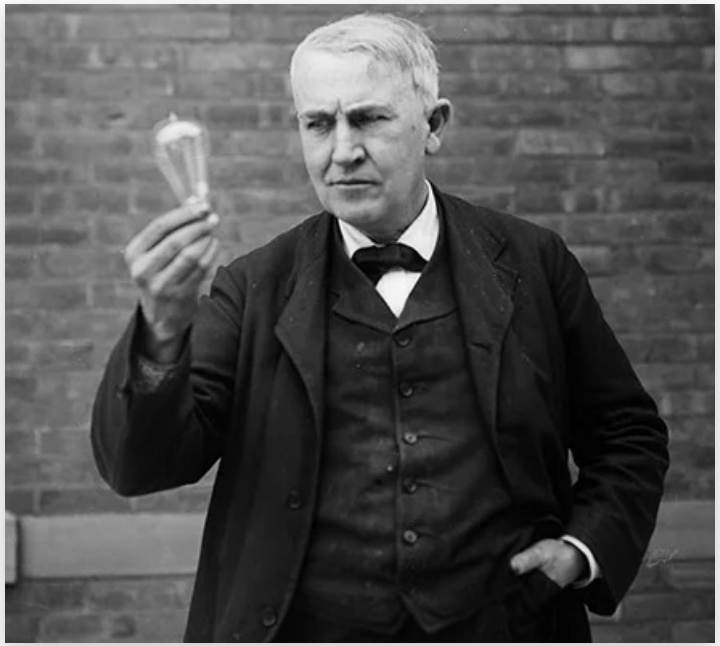



3D

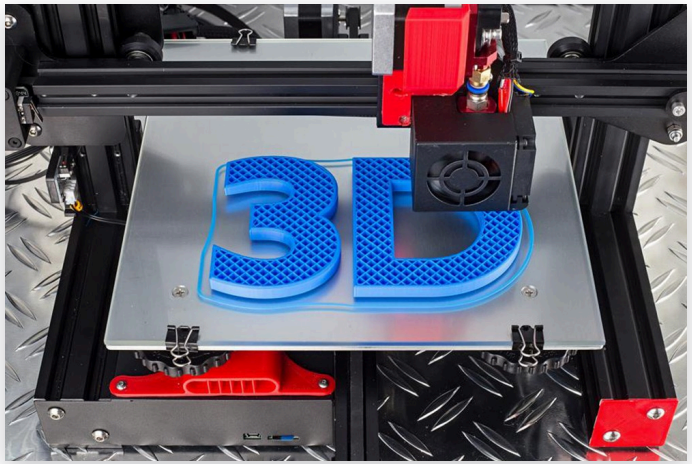




Light bulb



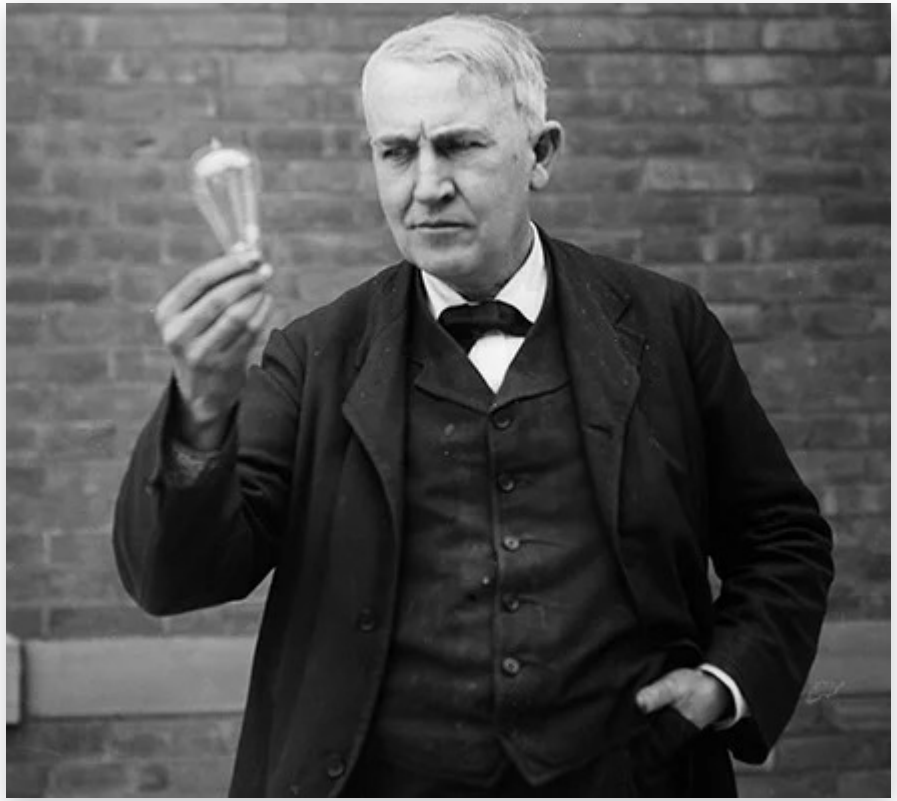




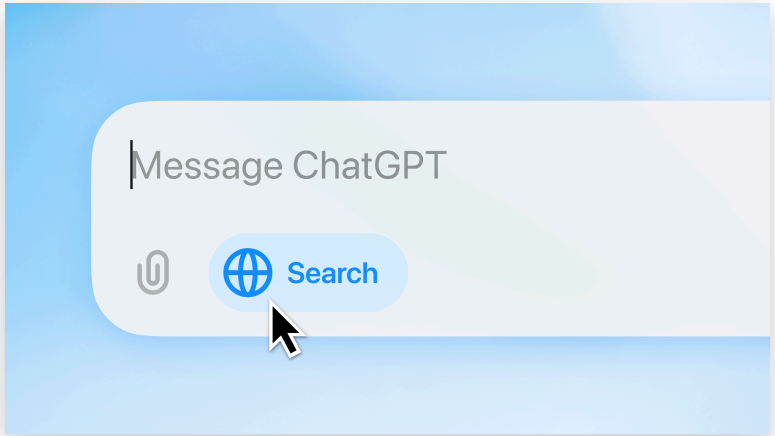
Light bulb changed human world

more than you'd think

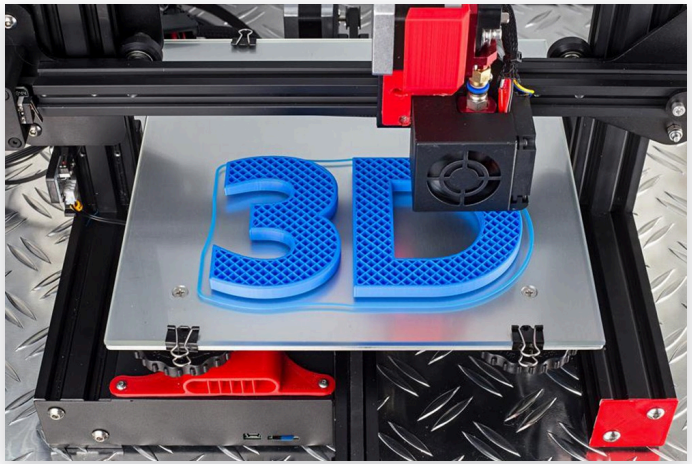
- 4–6 more hours a day
- The 24-hour economy
- Evening entertainment culture

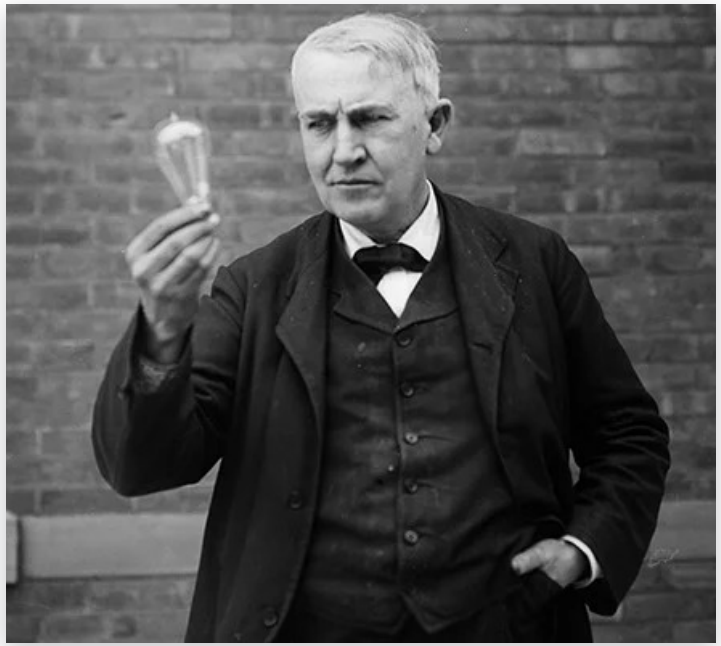



Google
Duo
Assistant
Play Store


Message ChatGPT
Search



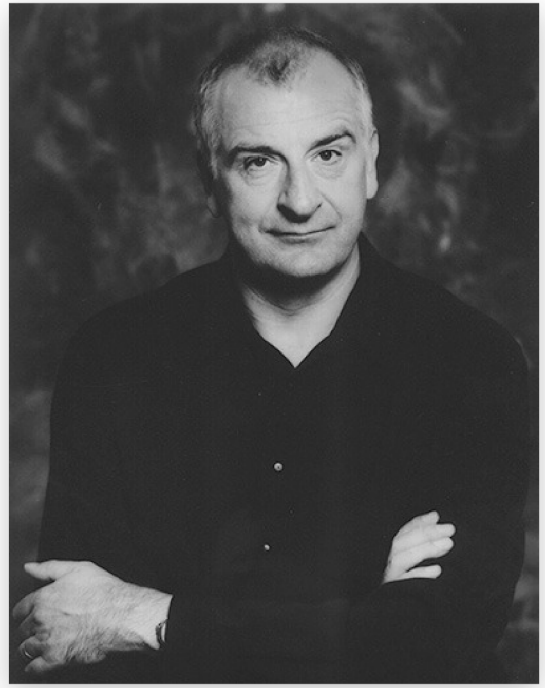

3D



“”

Anything that is in the world when you’re born is normal and ordinary and is just a natural part of the way the world works.

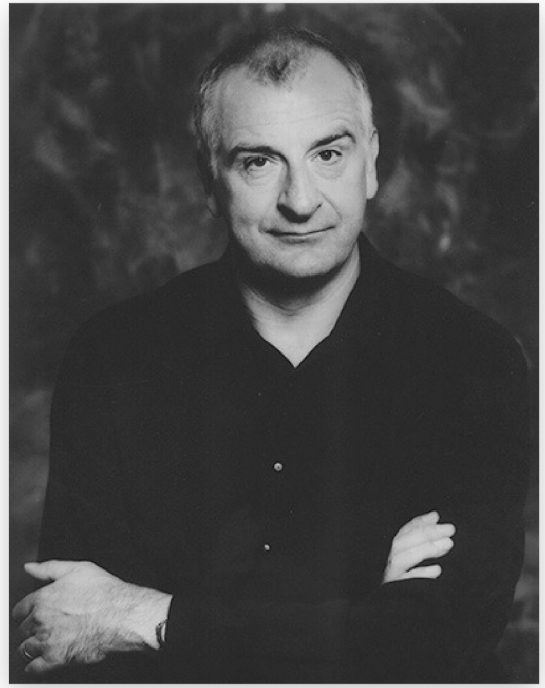
“The Salmon of Doubt”, Douglas Adams



“”

Anything that’s invented between when you’re fifteen and thirty-five is new and exciting and revolutionary and you can probably get a career in it.

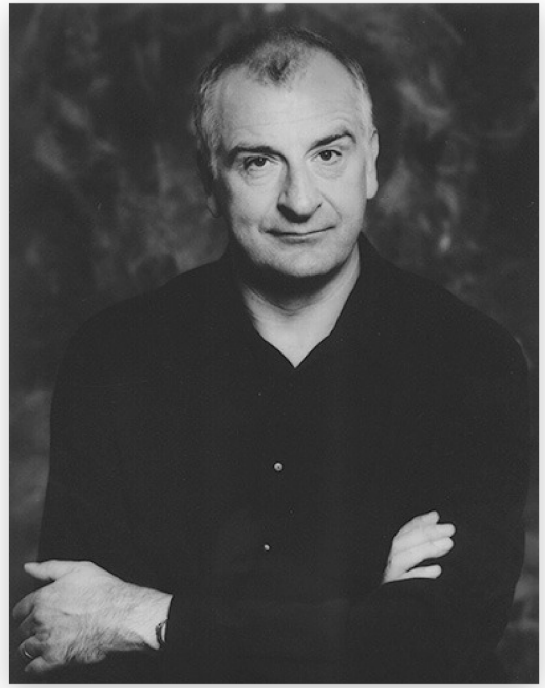
“The Salmon of Doubt”, Douglas Adams



“”

Anything invented after you’re thirty-five is against the natural order of things.

“The Salmon of Doubt”, Douglas Adams



My age



My age

6y old — *12y old* — *19y old* — *36y old*

Pagers



↑

This was normal

My age

6y old — *12y old* — *19y old* — *36y old*

Pagers



This was normal

Wireless phone



Innovative 🤯

My age

6y old — *12y old* — *19y old* — *36y old*

Pagers



This was normal

Wireless phone



Innovative 🤯

Smart phone



super cool!

My age

6y old — *12y old* — *19y old* — *36y old*

Pagers



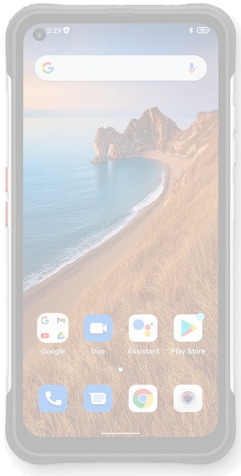
This was normal

Wireless phone



Innovative 🤯

Smart phone



super cool!

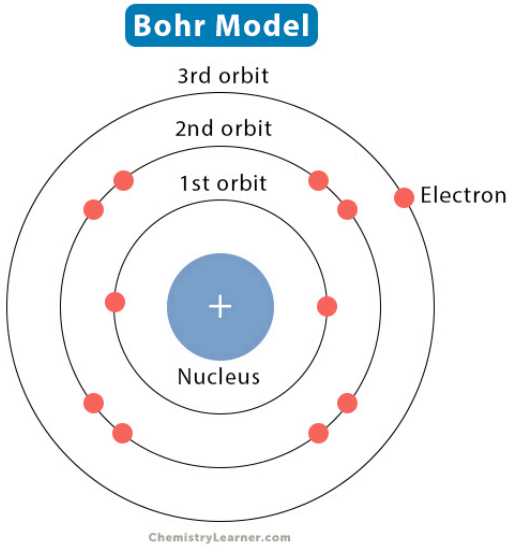
Smart glass



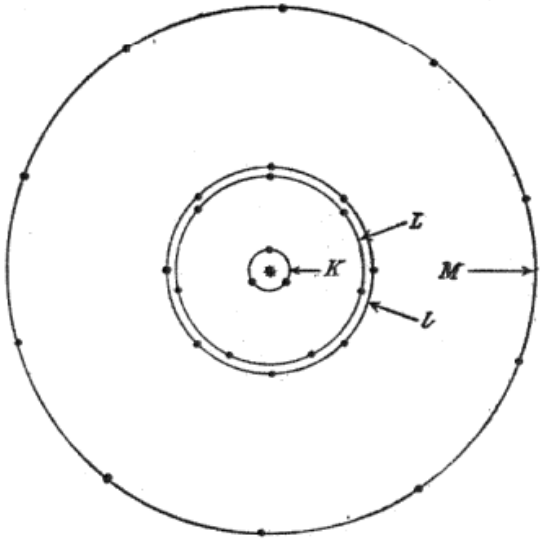
Goes against natural order


NOKIA
NOKIA

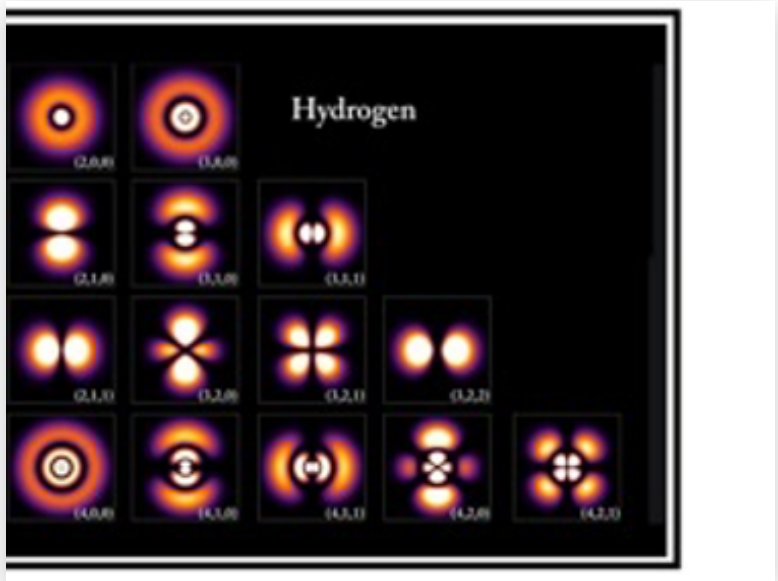

cingular
9:41 AM
SMS
Text
Tuesday
9
Calendar
Photos
Camera
Calculator
Stocks
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Maps
73°
Weather
Notes
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Settings
Phone
Mail
Web
iPod
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Bohr's model (1913)

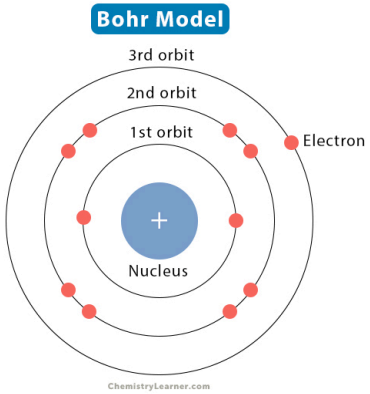


Vegard's model (1918)

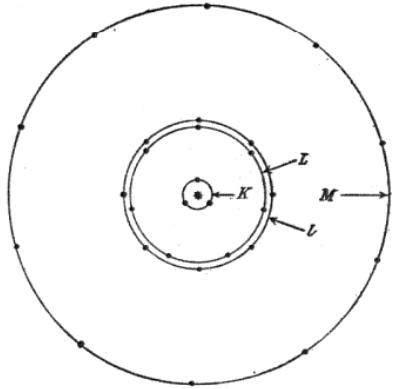


Schrödinger's model (1926)

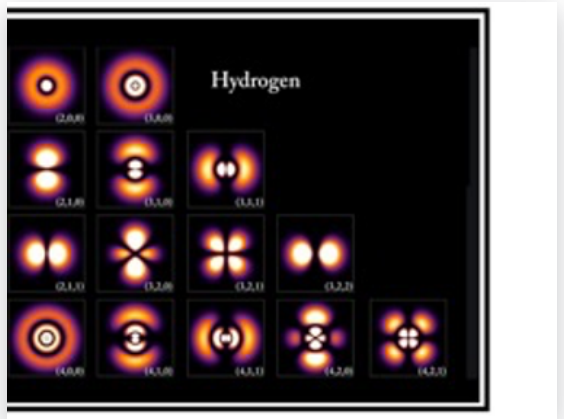
How do we quantify disruptiveness?



Bohr's model (1913)



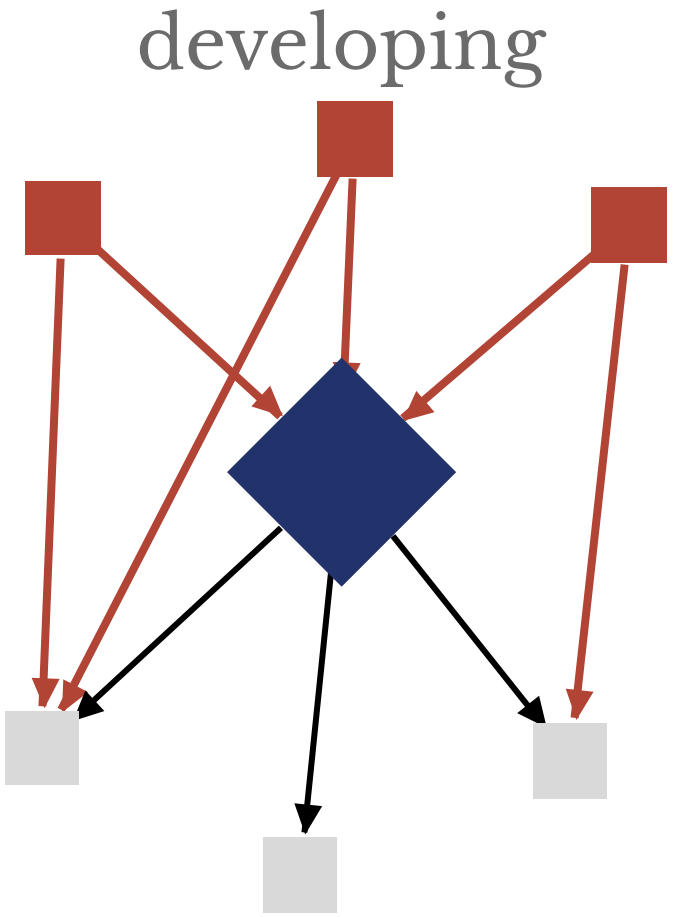
Vegard's model (1918)

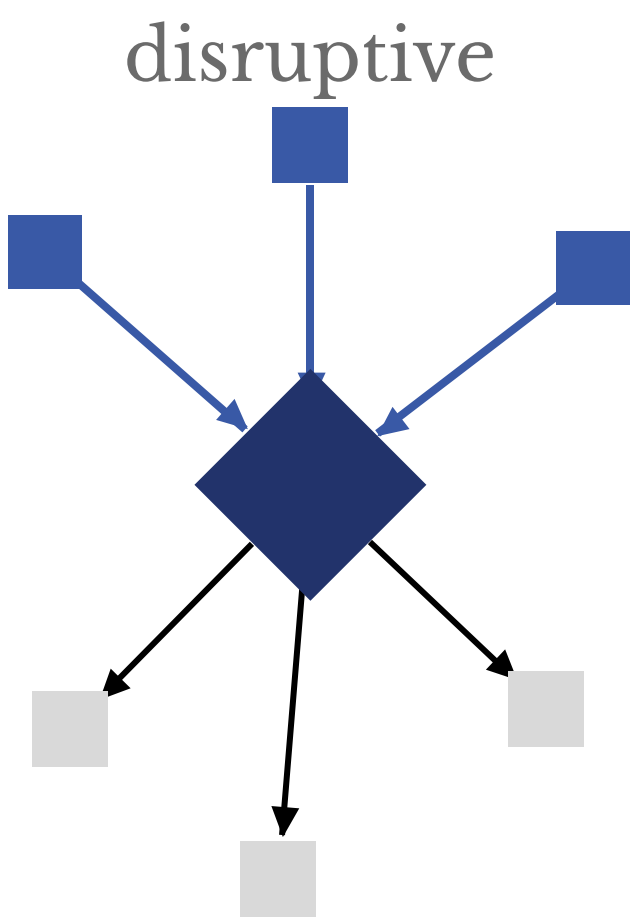


Schrödinger's model (1926)

Disruption index (Funk and Owen-Smith 2017)

Based on citation networks

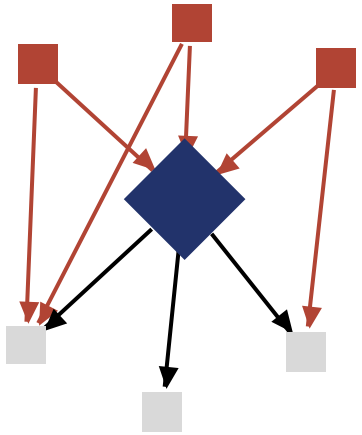




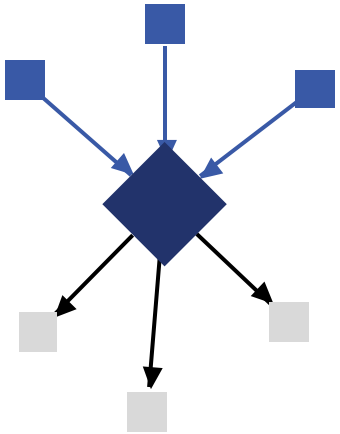
Disruption index (Funk and Owen-Smith 2017)

developing

disruptive

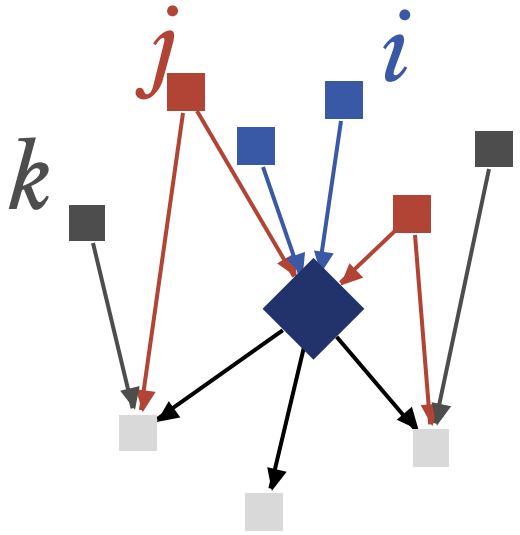




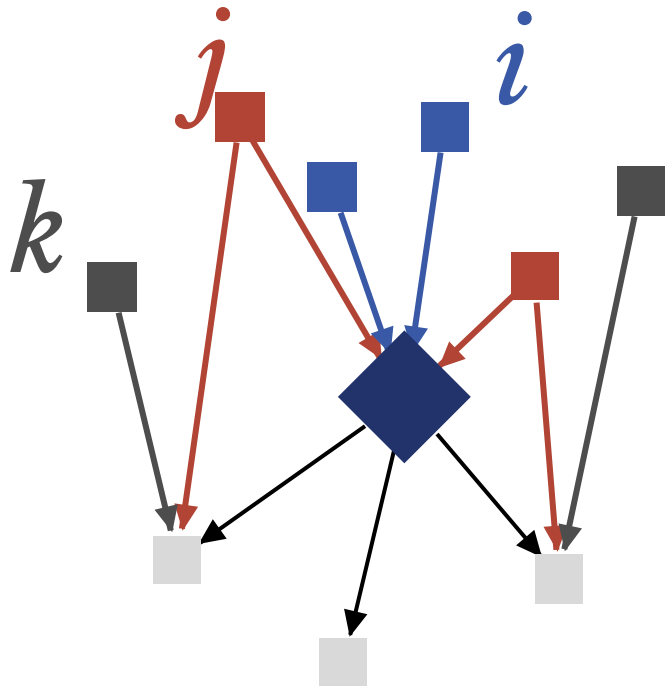


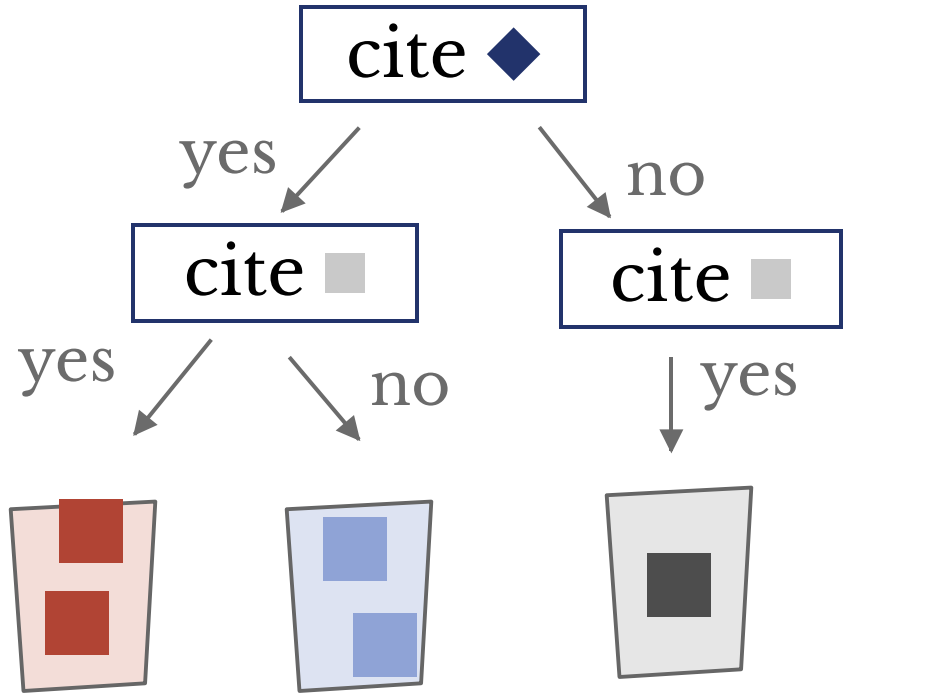
D=−1

D=1



Disruption index (Funk and Owen-Smith 2017)



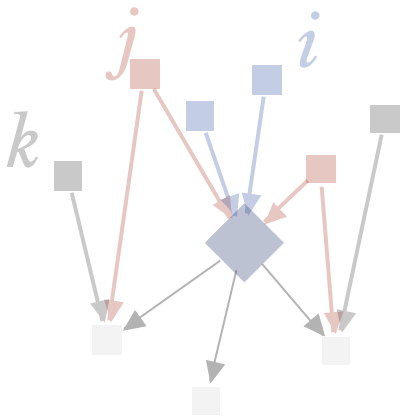


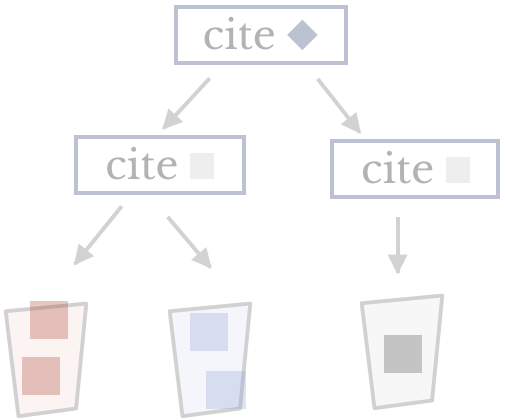
citing focal & its refs

citing the focal

citing its refs but focal

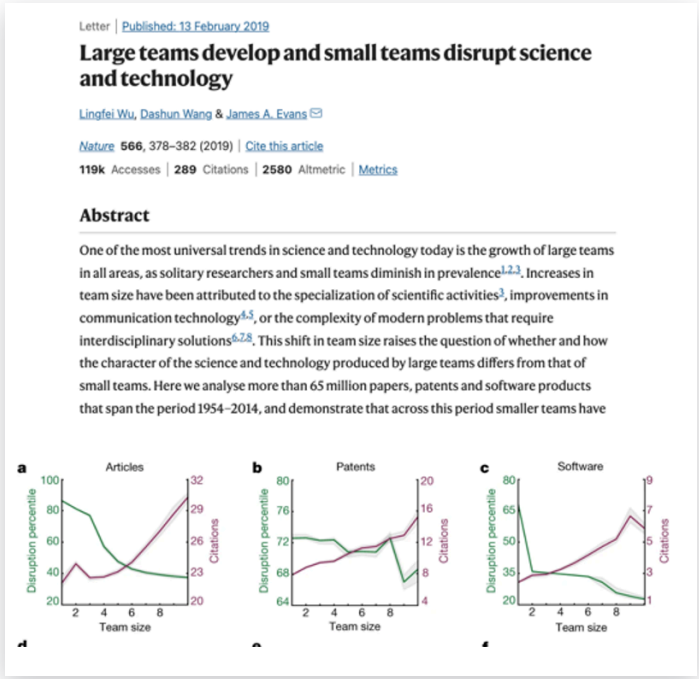
Disruption index (Funk and Owen-Smith 2017)





$$D_{\blacklozenge} = \frac{n_i - n_j}{n_i + n_j + n_k}$$

counts



Letter | Published: 13 February 2019

Large teams develop and small teams disrupt science and technology

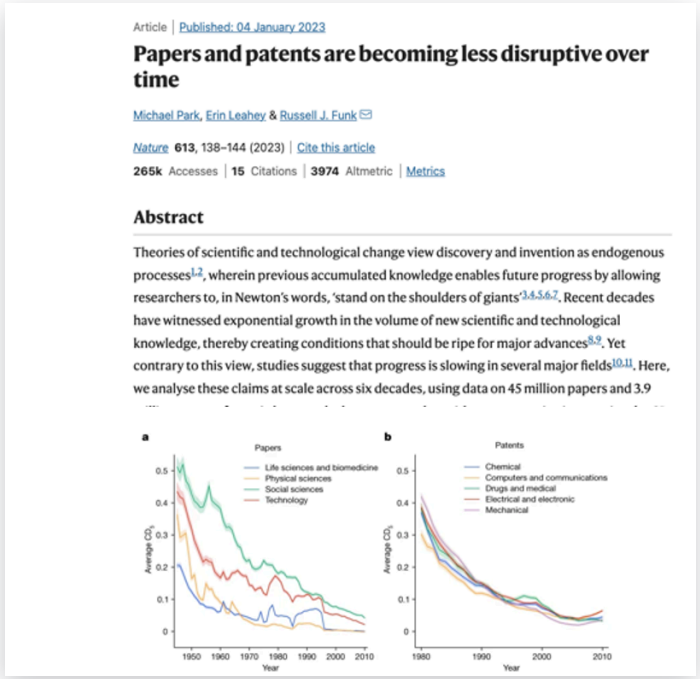
Lingfei Wu, Dashun Wang & James A. Evans

Nature **566**, 378–382 (2019) | Cite this article

119k Accesses | **289** Citations | **2580** Altmetric | Metrics

Abstract

One of the most universal trends in science and technology today is the growth of large teams in all areas, as solitary researchers and small teams diminish in prevalence[1,2,3]. Increases in team size have been attributed to the specialization of scientific activities[3], improvements in communication technology[4,5], or the complexity of modern problems that require interdisciplinary solutions[6,7,8]. This shift in team size raises the question of whether and how the character of the science and technology produced by large teams differs from that of small teams. Here we analyse more than 65 million papers, patents and software products that span the period 1954–2014, and demonstrate that across this period smaller teams have



Article | Published: 04 January 2023

Papers and patents are becoming less disruptive over time

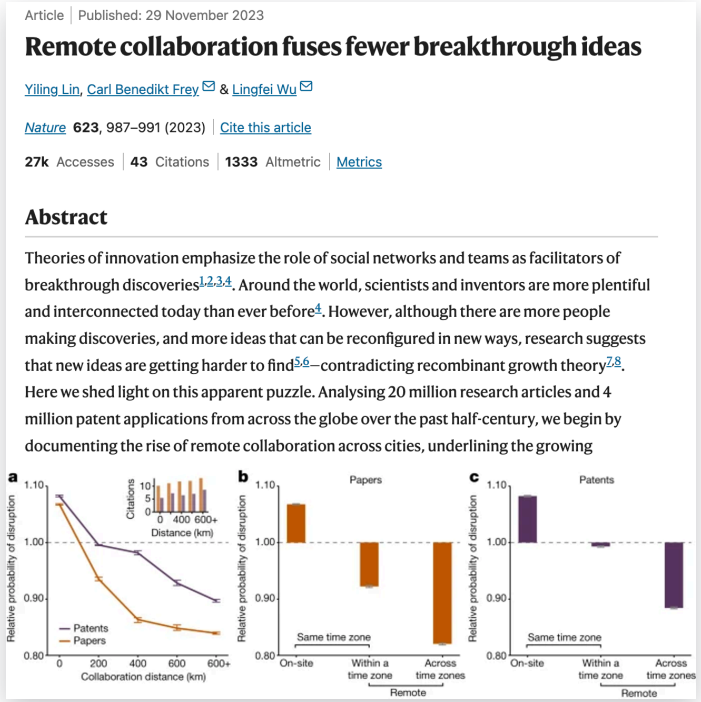
Michael Park, Erin Leahey & Russell J. Funk

Nature **613**, 138–144 (2023) | Cite this article

265k Accesses | **15** Citations | **3974** Altmetric | Metrics

Abstract

Theories of scientific and technological change view discovery and invention as endogenous processes[1,2], wherein previous accumulated knowledge enables future progress by allowing researchers to, in Newton's words, 'stand on the shoulders of giants'[3,4,5,6,7]. Recent decades have witnessed exponential growth in the volume of new scientific and technological knowledge, thereby creating conditions that should be ripe for major advances[8,9]. Yet contrary to this view, studies suggest that progress is slowing in several major fields[10,11]. Here, we analyse these claims at scale across six decades, using data on 45 million papers and 3.9



Article | Published: 29 November 2023

Remote collaboration fuses fewer breakthrough ideas

Yiling Lin, Carl Benedikt Frey & Lingfei Wu

Nature **623**, 987–991 (2023) | Cite this article

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Abstract

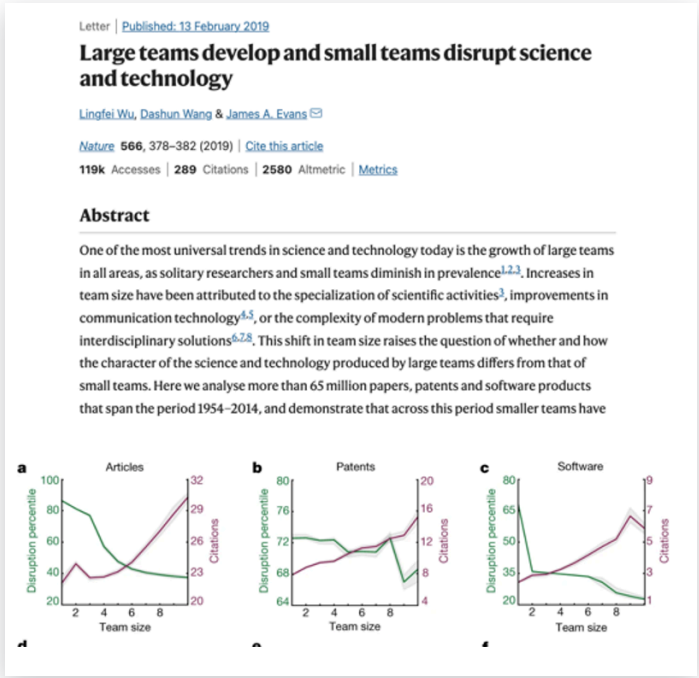
Theories of innovation emphasize the role of social networks and teams as facilitators of breakthrough discoveries[1,2,3,4]. Around the world, scientists and inventors are more plentiful and interconnected today than ever before[4]. However, although there are more people making discoveries, and more ideas that can be reconfigured in new ways, research suggests that new ideas are getting harder to find[5,6]—contradicting recombinant growth theory[7,8]. Here we shed light on this apparent puzzle. Analysing 20 million research articles and 4 million patent applications from across the globe over the past half-century, we begin by documenting the rise of remote collaboration across cities, underlining the growing

Team size

Temporal Trend

Modes of Communication

Wu, Wang & Evans (2019) *Nature* 566. Park, Leahy & Funk (2023) *Nature* 613. Lin, Frey & Wu (2023) *Nature* 623.



Letter | Published: 13 February 2019

Large teams develop and small teams disrupt science and technology

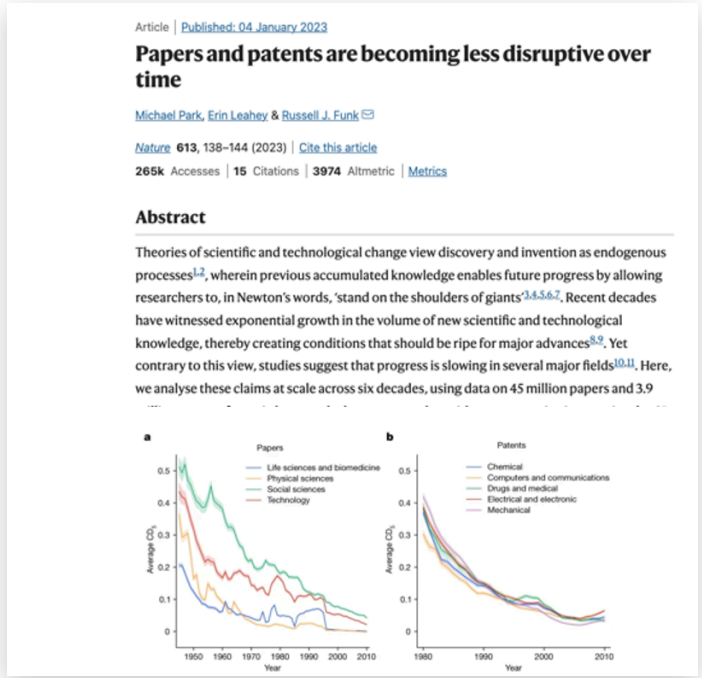
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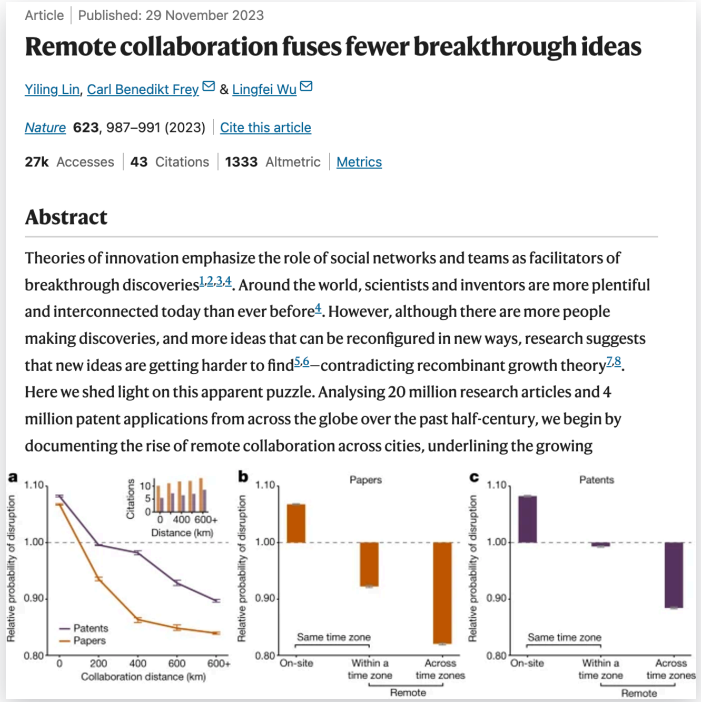
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Team size

Temporal Trend

Modes of Communication

BUT, the D-index fails for some well-known disruptive discoveries

- **Discovery**
  - 1964: Englert, Brout and Higgs predict the boson
- **Recognition**
  - 2013: Nobel Prize in Physics



- **Discovery**
  - 1964: Englert, Brout and Higgs predict the boson
- **Recognition**
  - 2013: Nobel Prize in Physics

Englert, F.,
& Brout, R. (1964)

D = 0.15
Top 2.8 %



- **Discovery**
  - 1964: Englert, Brout and Higgs predict the boson
- **Recognition**
  - 2013: Nobel Prize in Physics

Englert, F.,
& Brout, R. (1964)

D = 0.15
Top 2.8 %



Higgs, P.W. (1964)

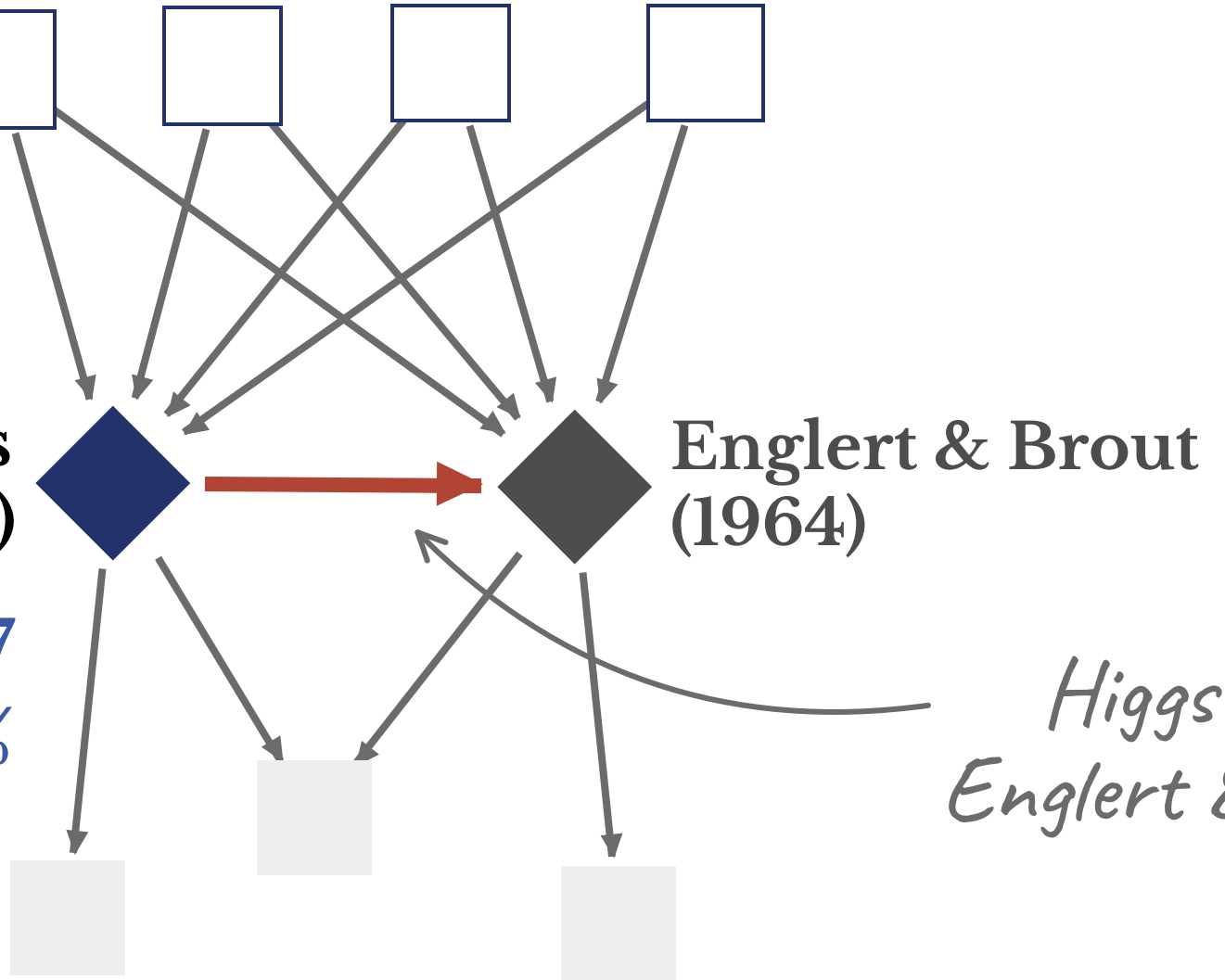
D = – 0.27
Bottom 0.1 %

What's happening?



What's happening?





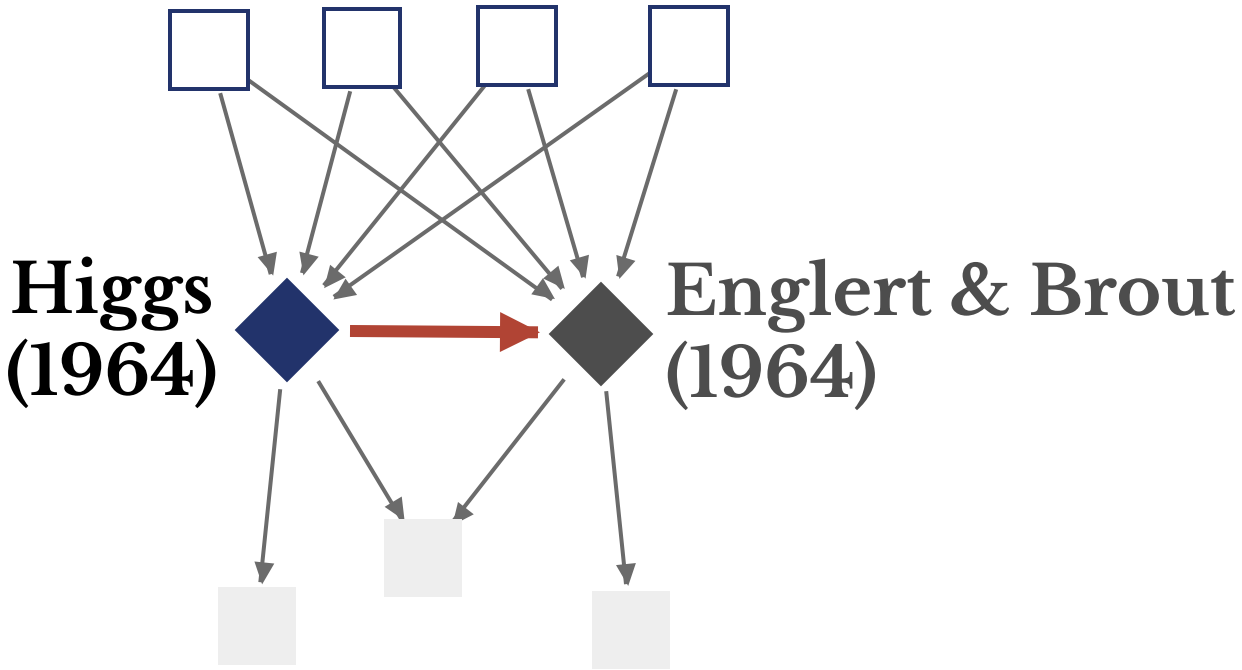
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Higgs cites
Englert & Brout.

What's happening?

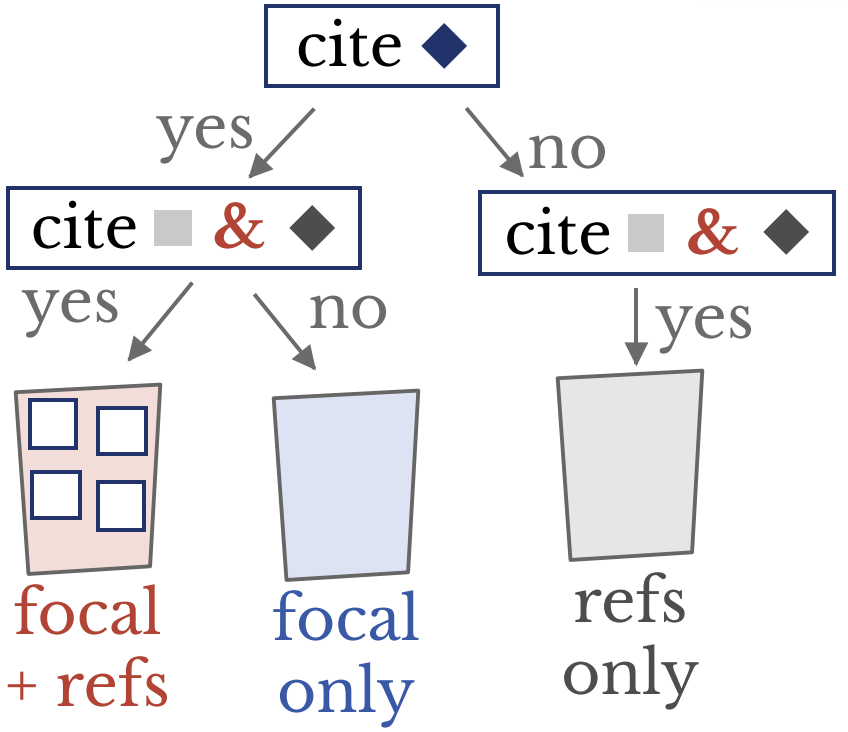
All four papers cite both.





D = – 0.27

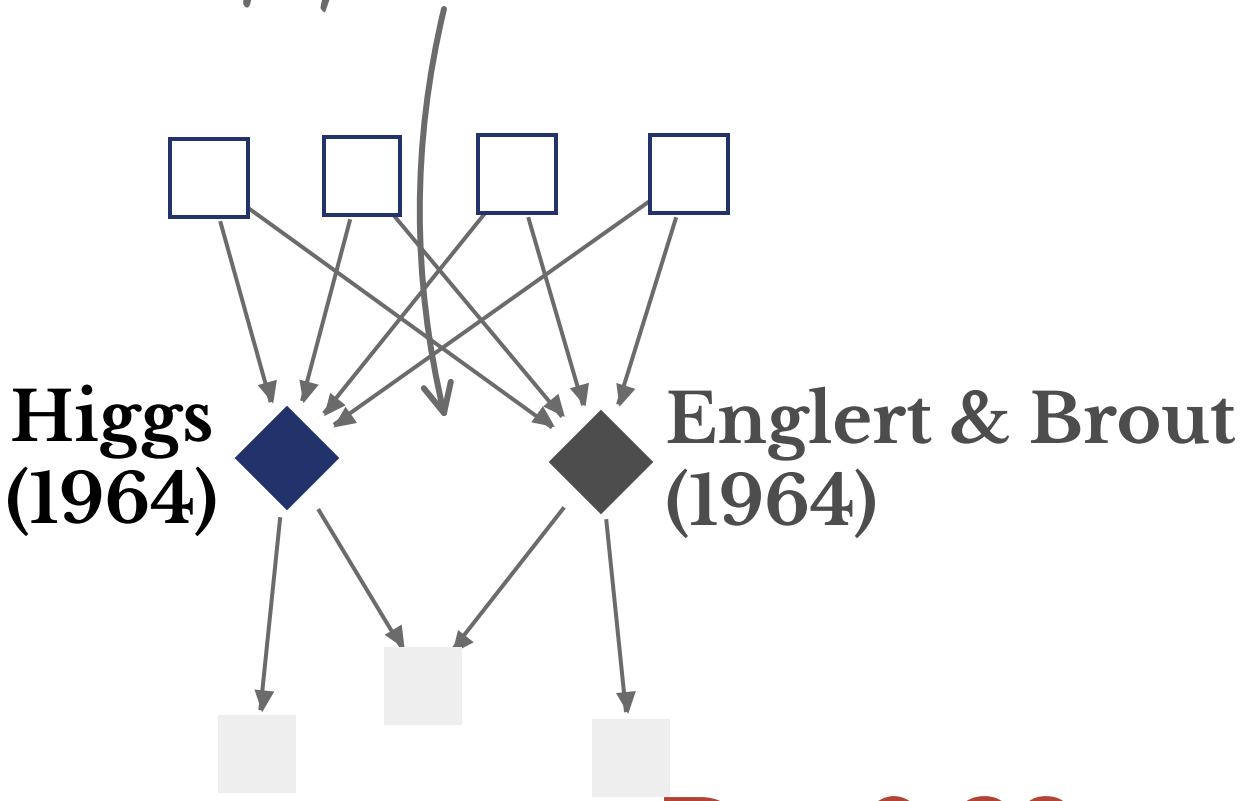
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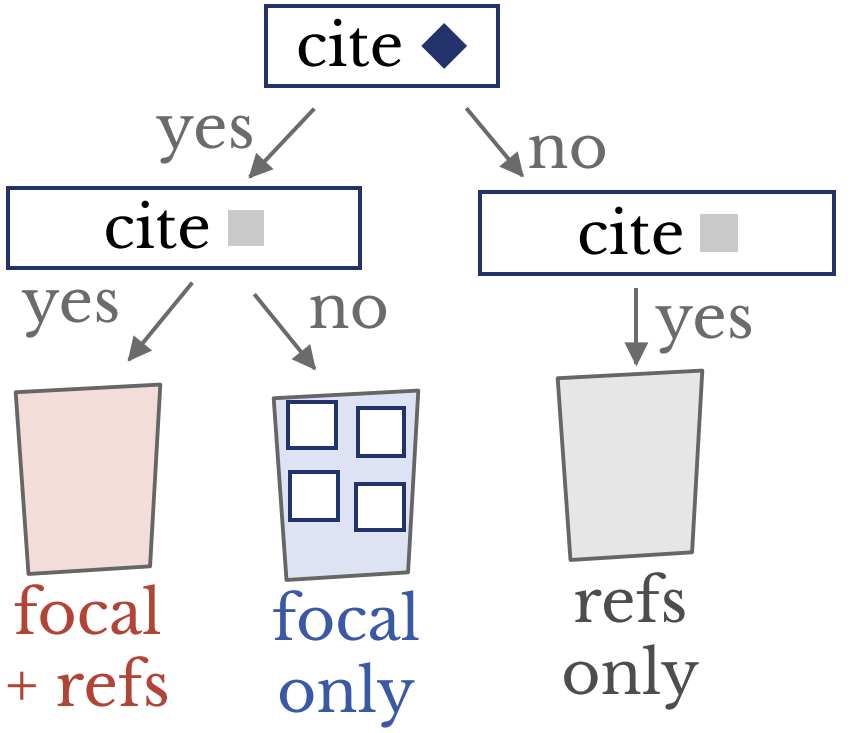


What's happening?

One citation removed.

All four papers switch class.



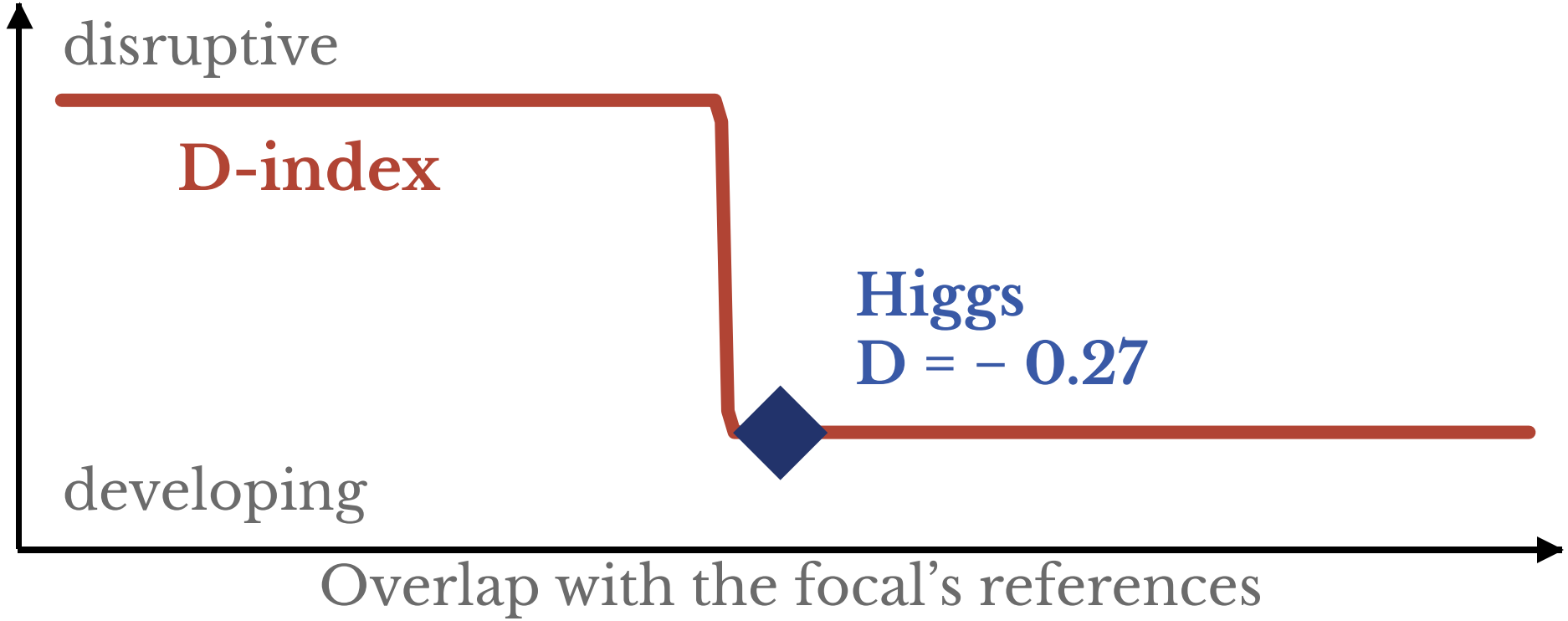


D = – 0.27
Bottom 0.1 %

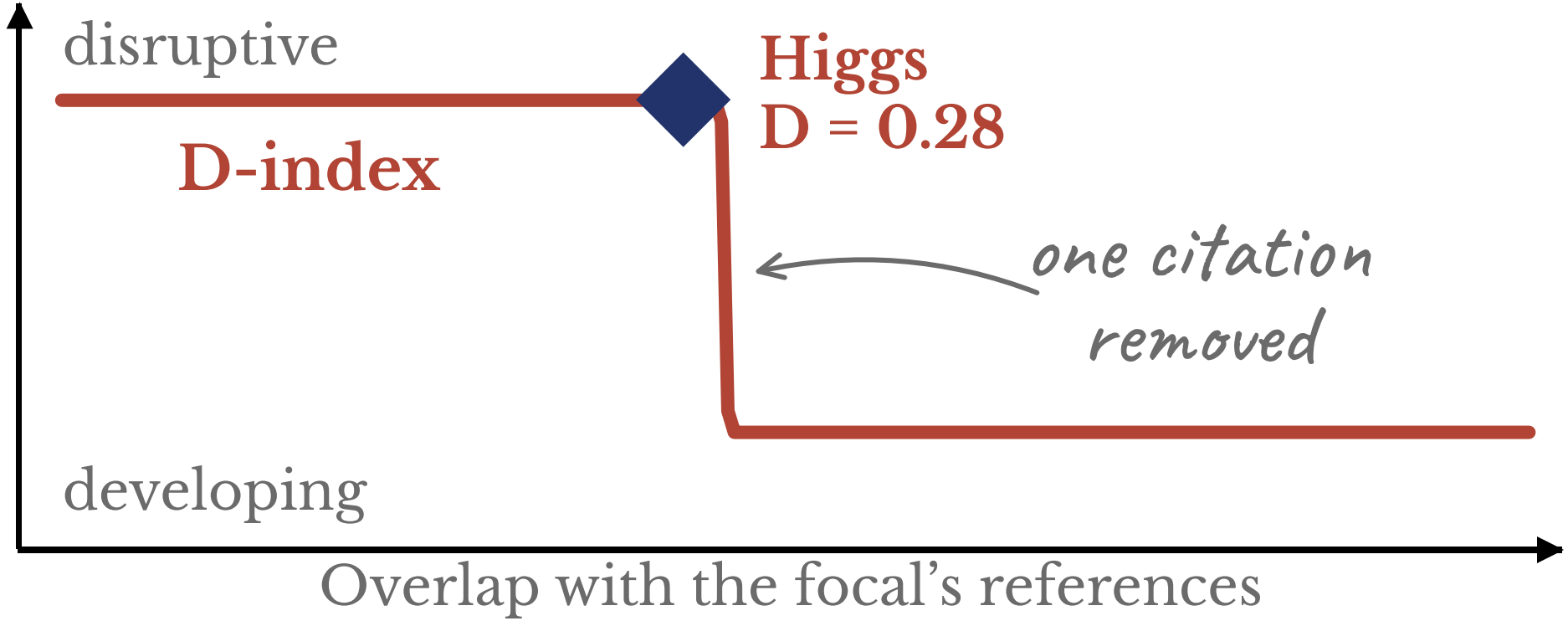
→

D = 0.28
Top 1.3 %

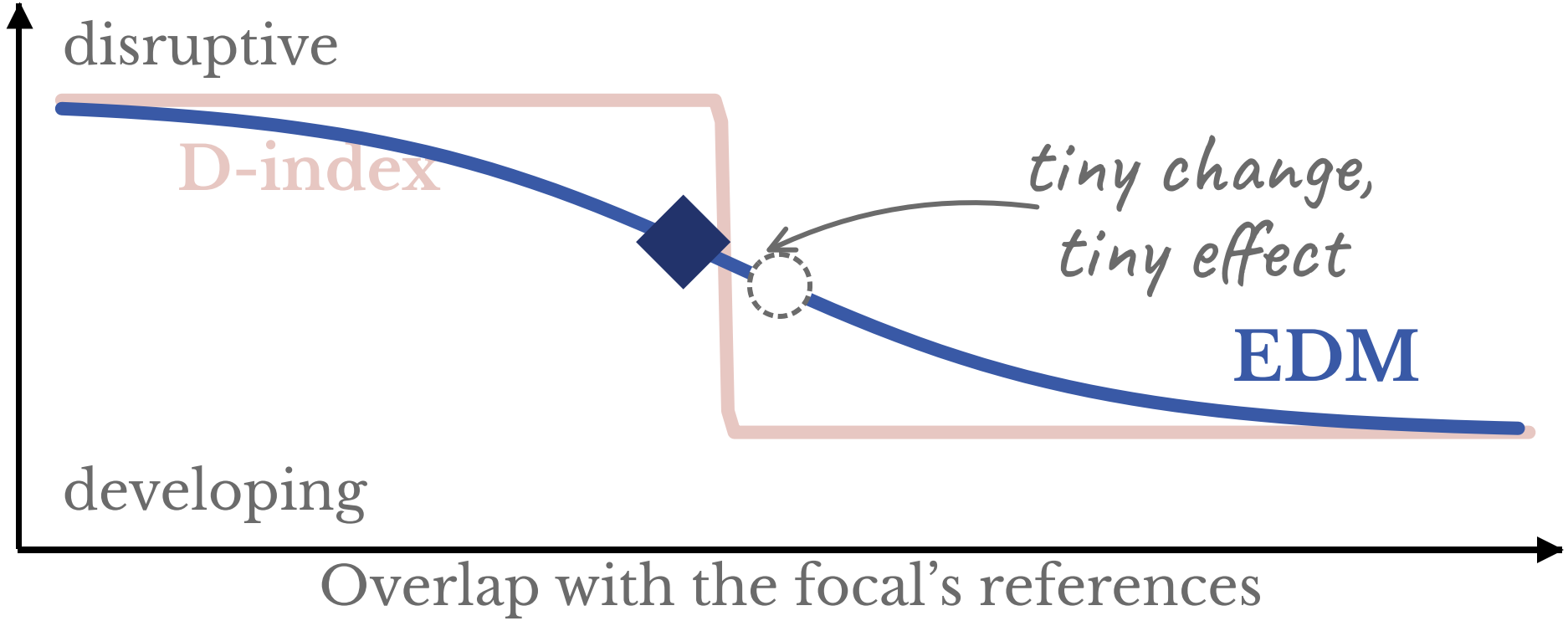
Citations are "*rounded*" into classes.



Citations are "*rounded*" into classes.

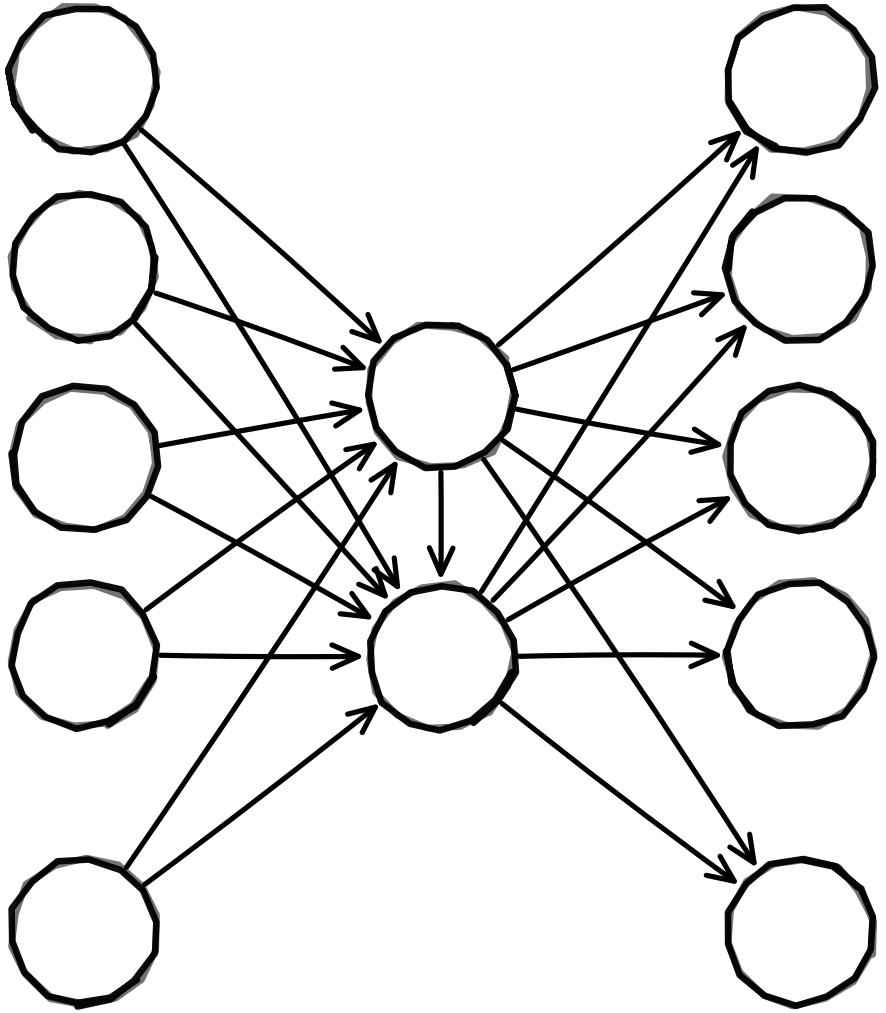


Citations are "*rounded*" into classes.

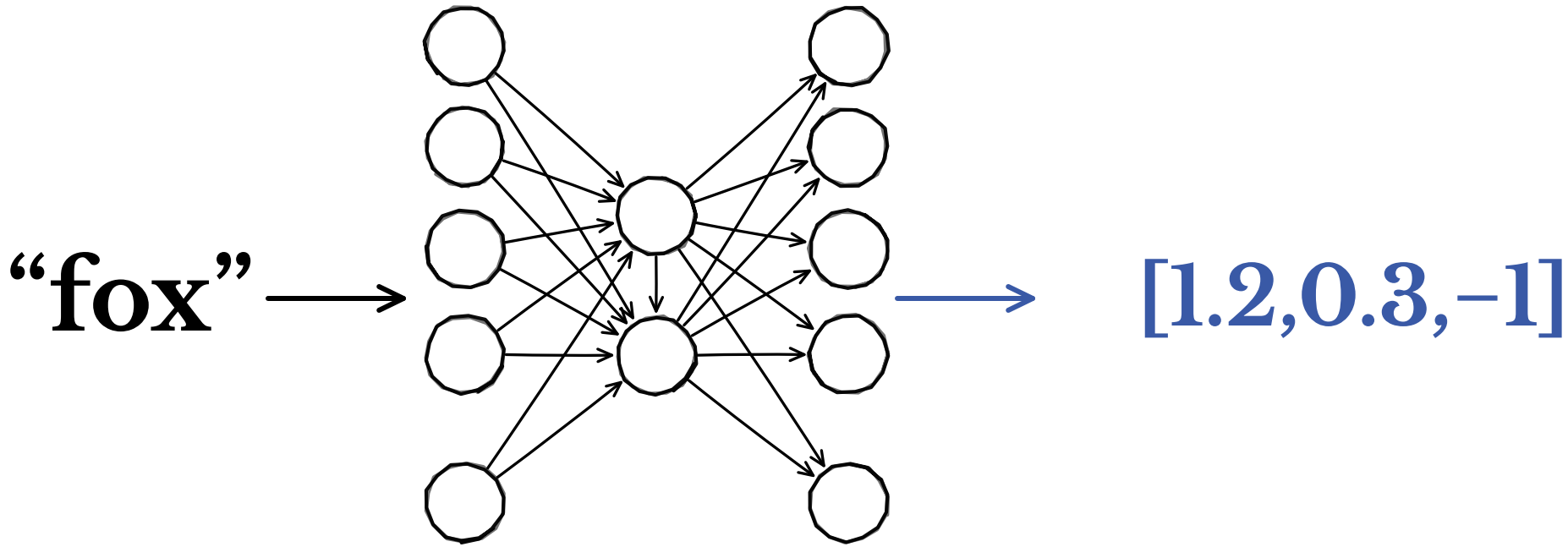


We preserve continuity by abandoning class rounding.

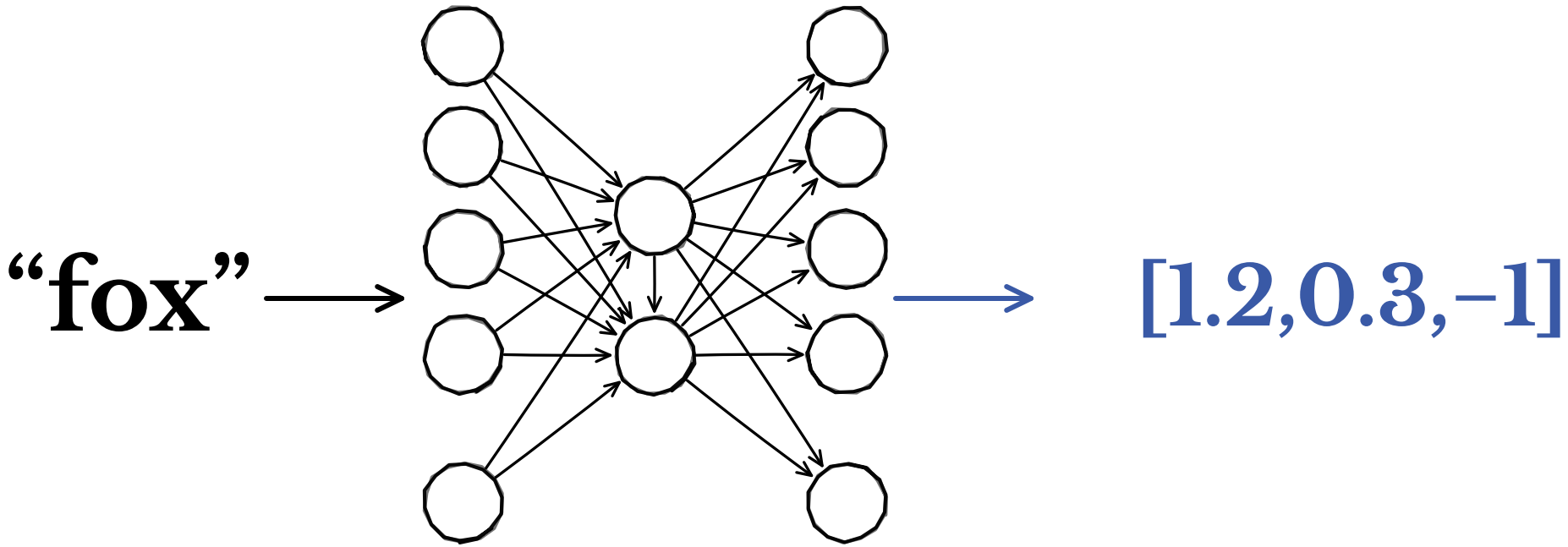
word2vec



word2vec (Mikolov et al. 2013)



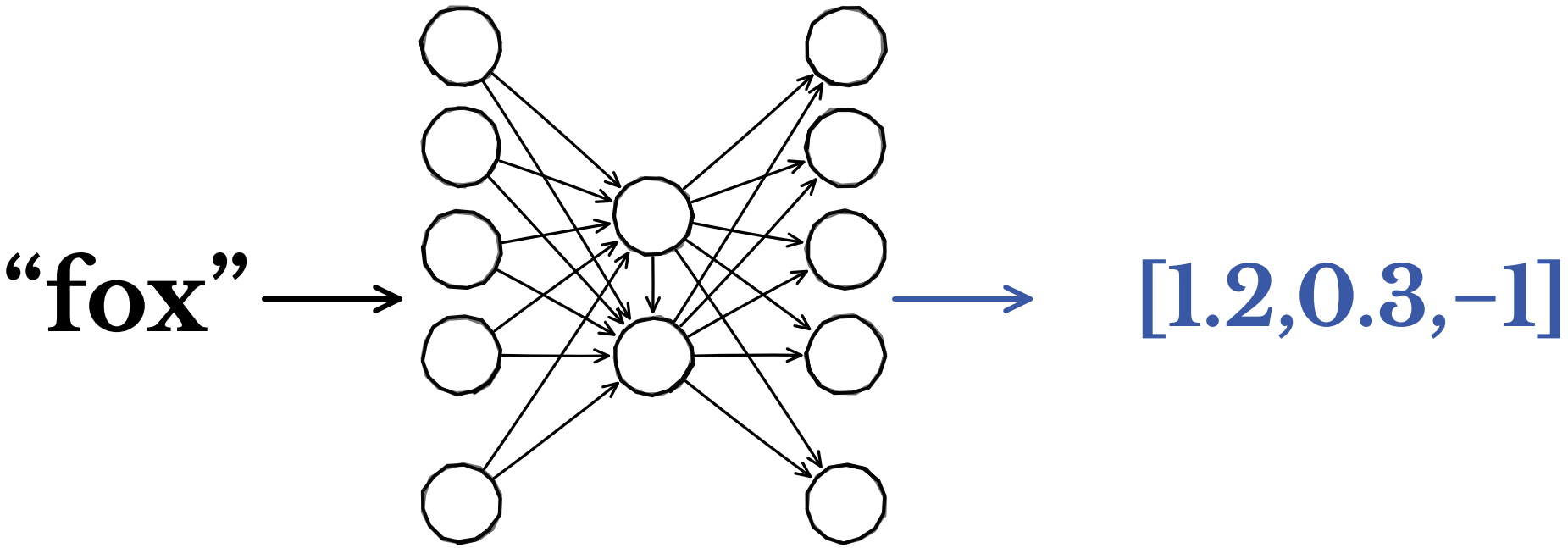
word2vec (Mikolov et al. 2013)



Learns by metonymy

Words that appear together sit close in the vector space.

word2vec (Mikolov et al. 2013)



Learns by metonymy

Words that appear together sit close in the vector space.

Surprisingly effective. Paradigm shift in NLP

Learn by metonymy but captures metaphor.

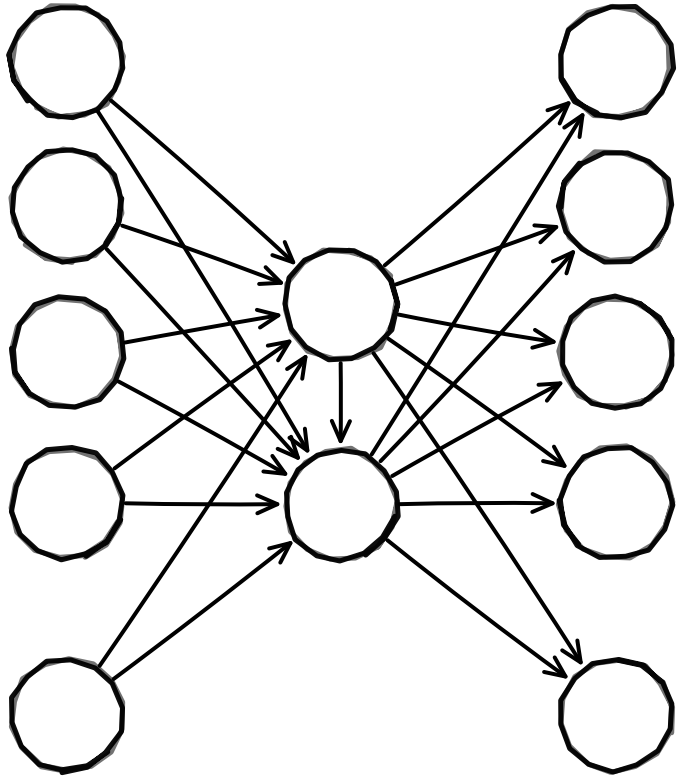
$$\overrightarrow{\text{“man”}} - \overrightarrow{\text{“king”}} + \overrightarrow{\text{“woman”}} = \overrightarrow{\text{“queen”}}$$

$$\overrightarrow{\text{“Madrid”}} - \overrightarrow{\text{“Spain”}} + \overrightarrow{\text{“Japan”}} = \overrightarrow{\text{“Tokyo”}}$$

Tested 10y> ago by myself

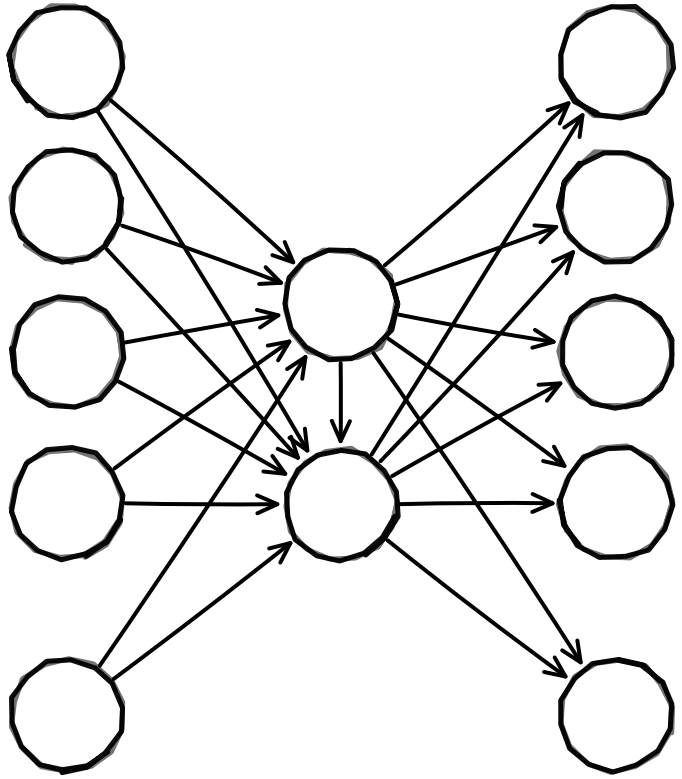
$$\overrightarrow{\text{“Taylor_Swift”}} - \overrightarrow{\text{“USA”}} + \overrightarrow{\text{“Japan”}} = \overrightarrow{\text{“Namie_Amuro”}}$$

How word2vec works



*word2vec has multiple variants. We focus on the skip-gram negative sampling word2vec.

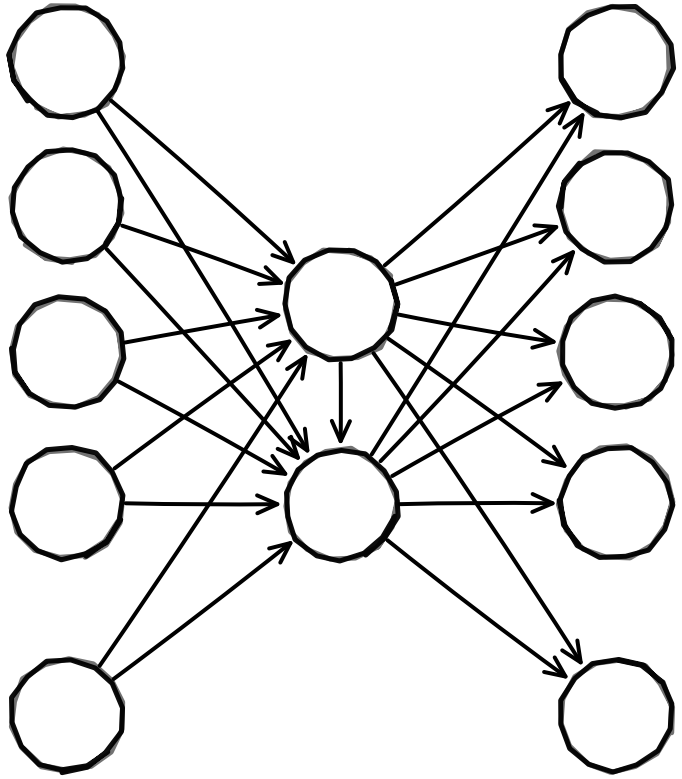
“Chance favors only the _____ mind.”



*word2vec has multiple variants. We focus on the skip-gram negative sampling word2vec.

“Chance favors only the _____ mind.”

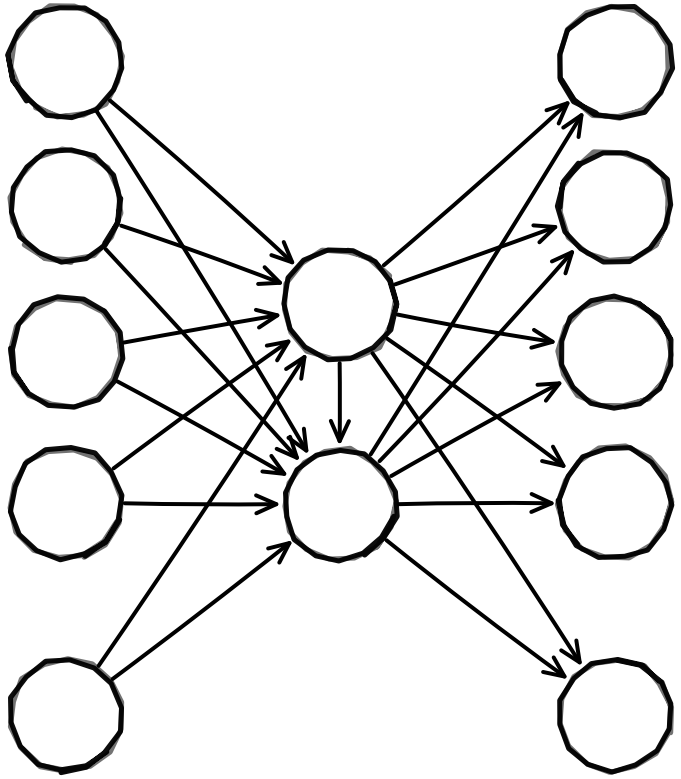
“favors”



*word2vec has multiple variants. We focus on the skip-gram negative sampling word2vec.

“Chance favors only the _____ mind.”

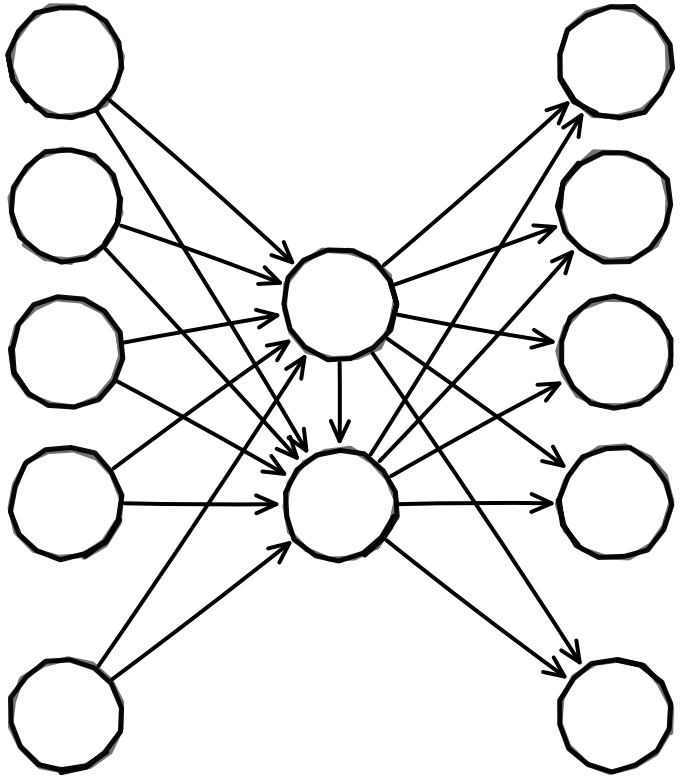
“favors”



“prepared”

*word2vec has multiple variants. We focus on the skip-gram negative sampling word2vec.

“favors”



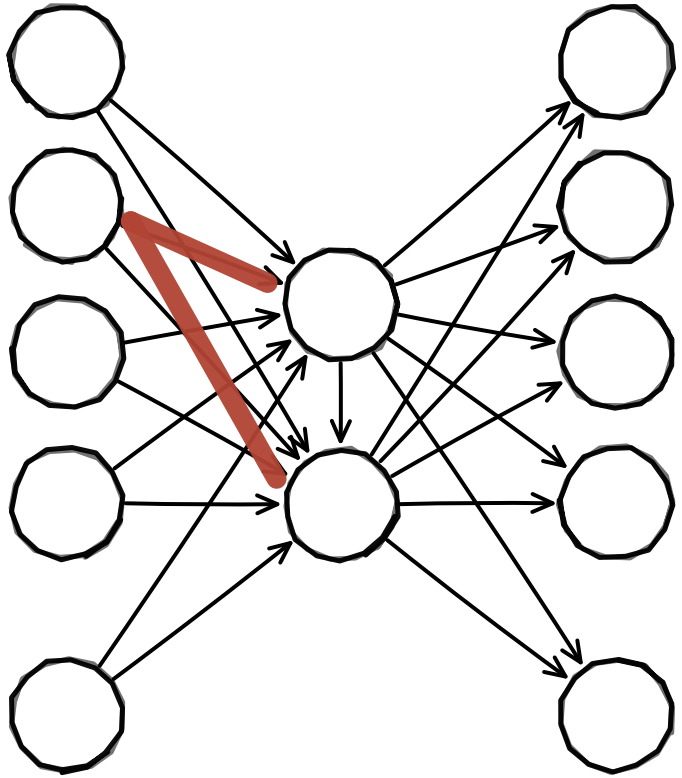
“prepared”

*word2vec has multiple variants. We focus on the skip-gram negative sampling word2vec.

$$q_{\text{favor}} = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} \begin{matrix} \text{N1} \\ \text{N2} \end{matrix}$$

query vector

“favors”



“prepared”

*word2vec has multiple variants. We focus on the skip-gram negative sampling word2vec.

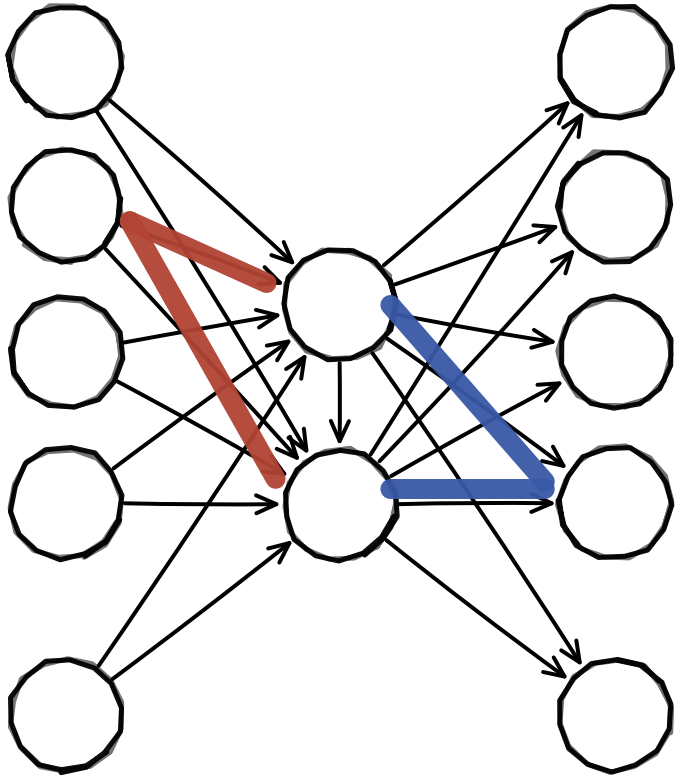
$$q_{\text{favor}} = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} \begin{matrix} \text{N1} \\ \text{N2} \end{matrix}$$

query vector

$$k_{\text{prepare}} = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix} \begin{matrix} \text{N1} \\ \text{N2} \end{matrix}$$

key vector

“favors”



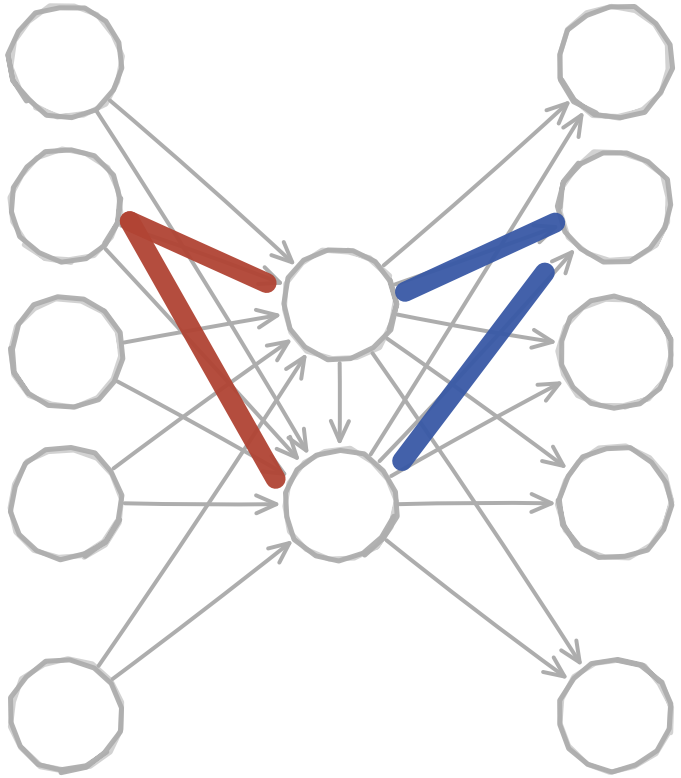
“prepared”

*word2vec has multiple variants. We focus on the skip-gram negative sampling word2vec.

A word has two vectors

Query

"favors"



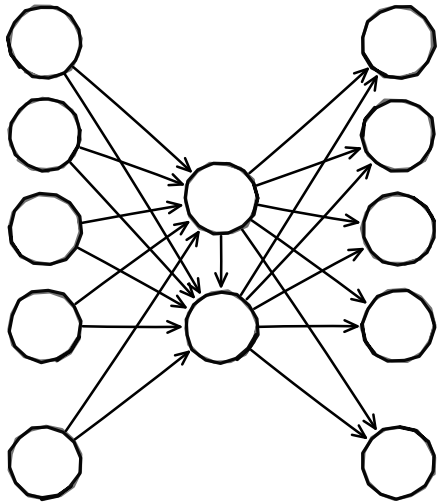
"favors"

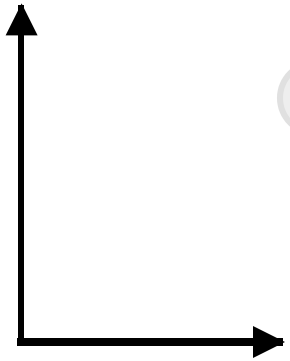
"prepared"

Key

Trained by negation.

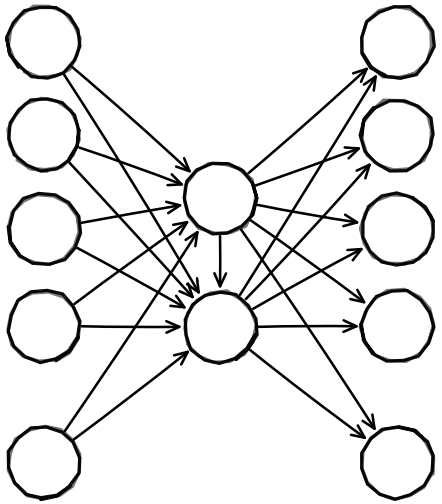
"... favors only the prepared"

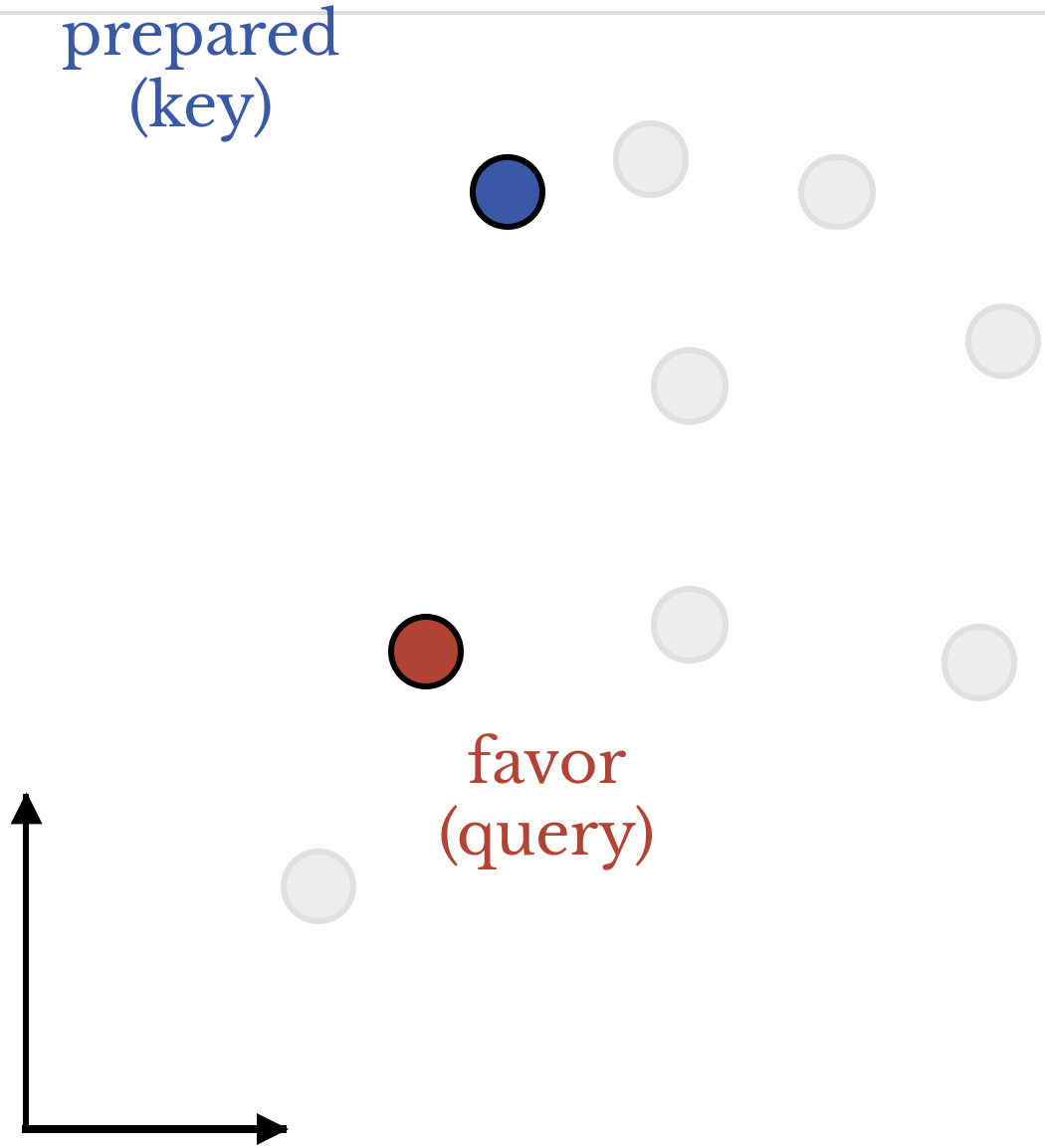




Trained by negation.

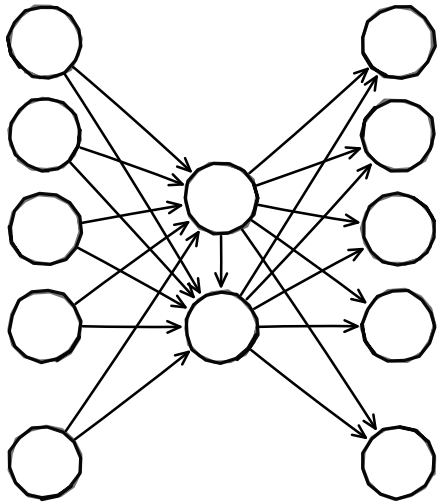
"... favors only the prepared"

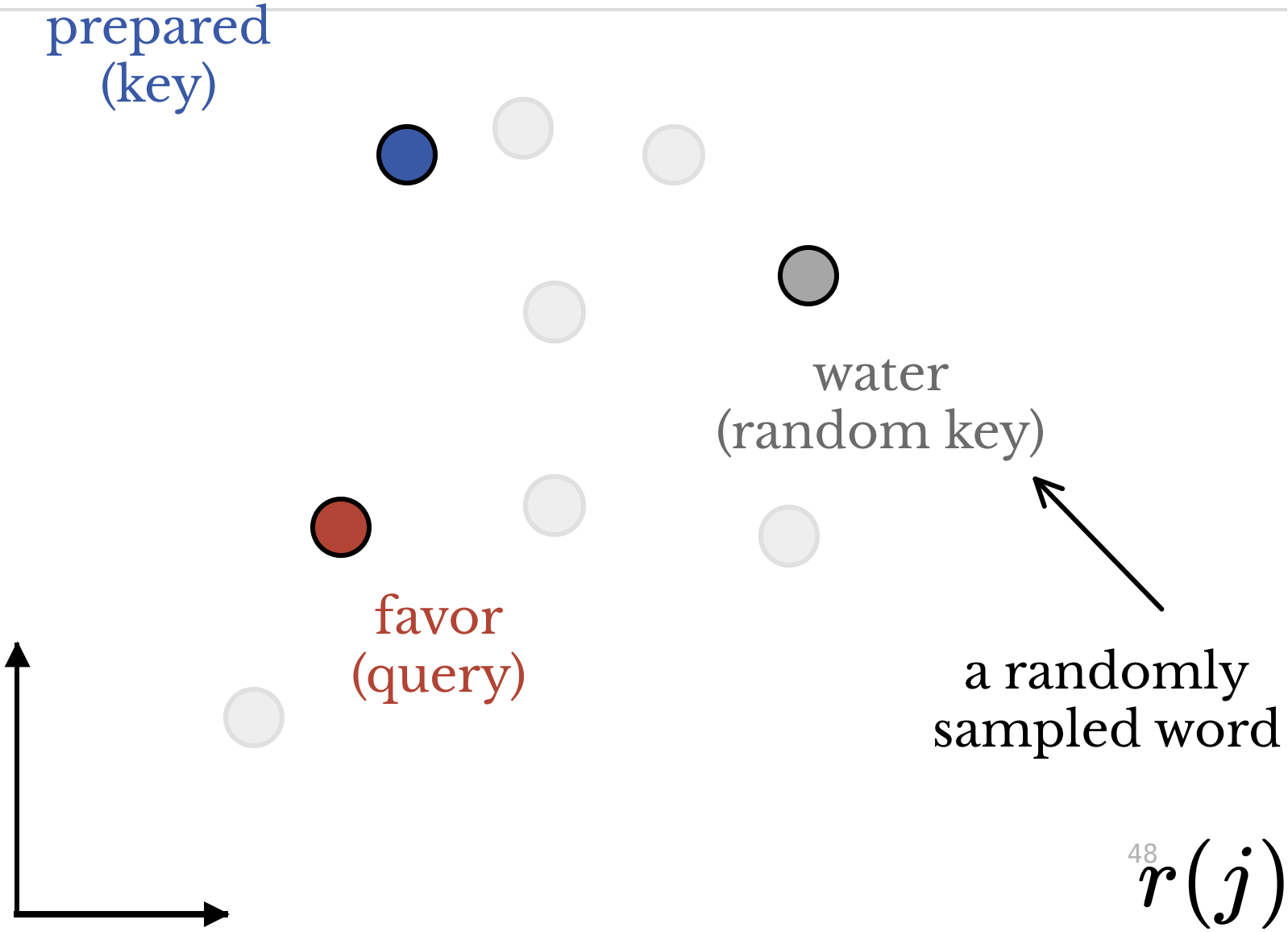




Trained by negation.

"... favors only the prepared"

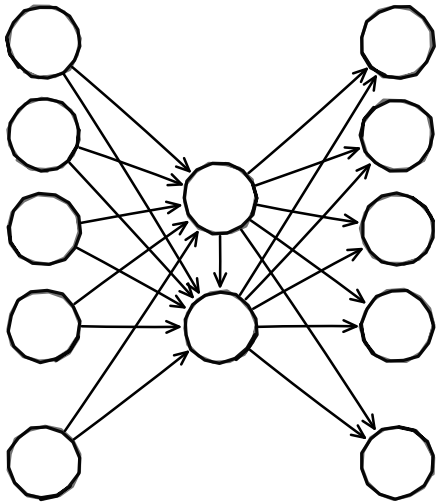


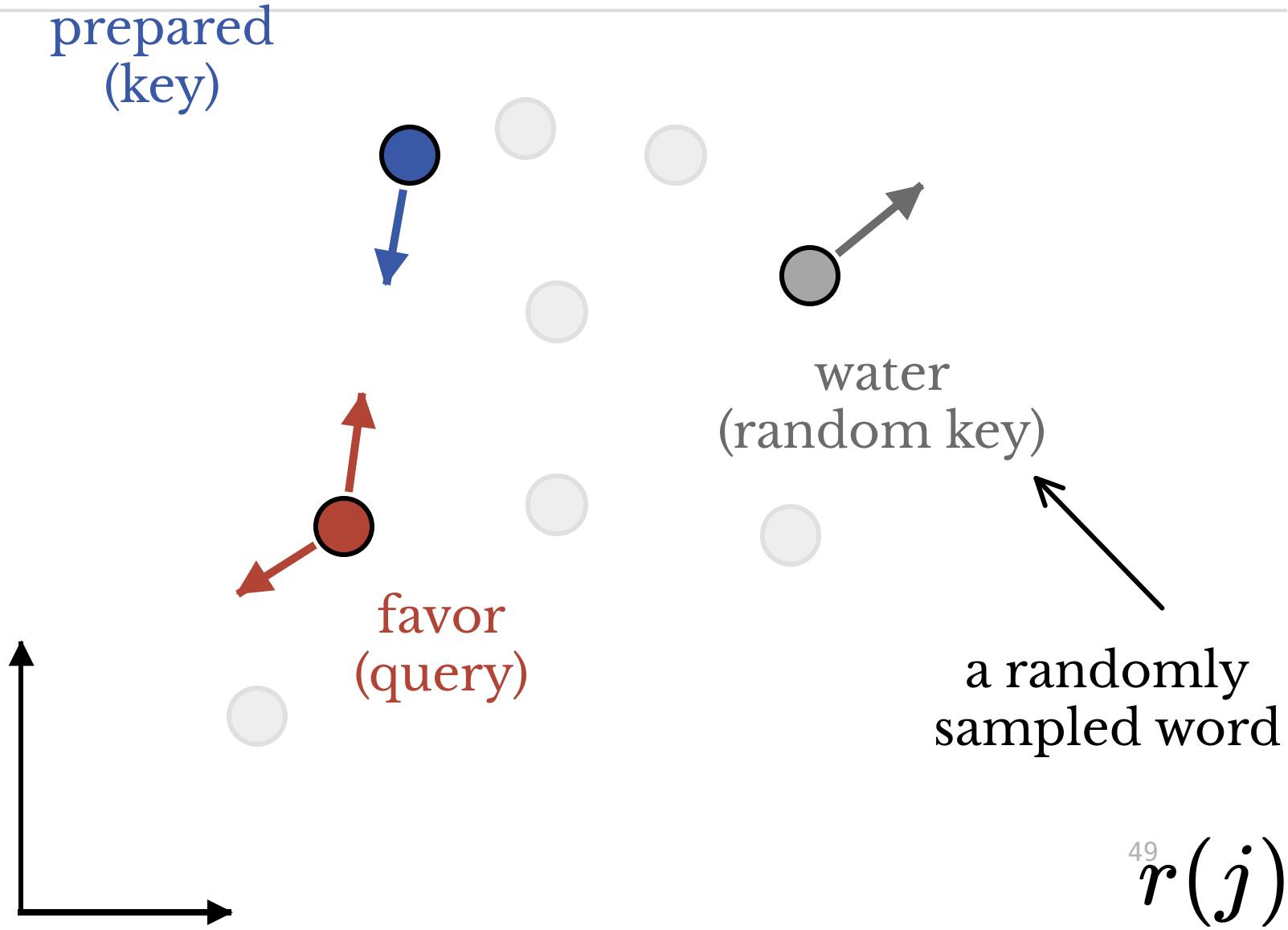


$r(j)$

Trained by negation.

"... favors only the prepared"





$r(j)$

query	key	dot sim (x)	genuine? (y)
favor 	prepared 	$q_{\text{favor}}^{\top} k_{\text{prepare}}$	Yes (y=1)
favor 	water 	$q_{\text{favor}}^{\top} k_{\text{water}}$	No (y=0)

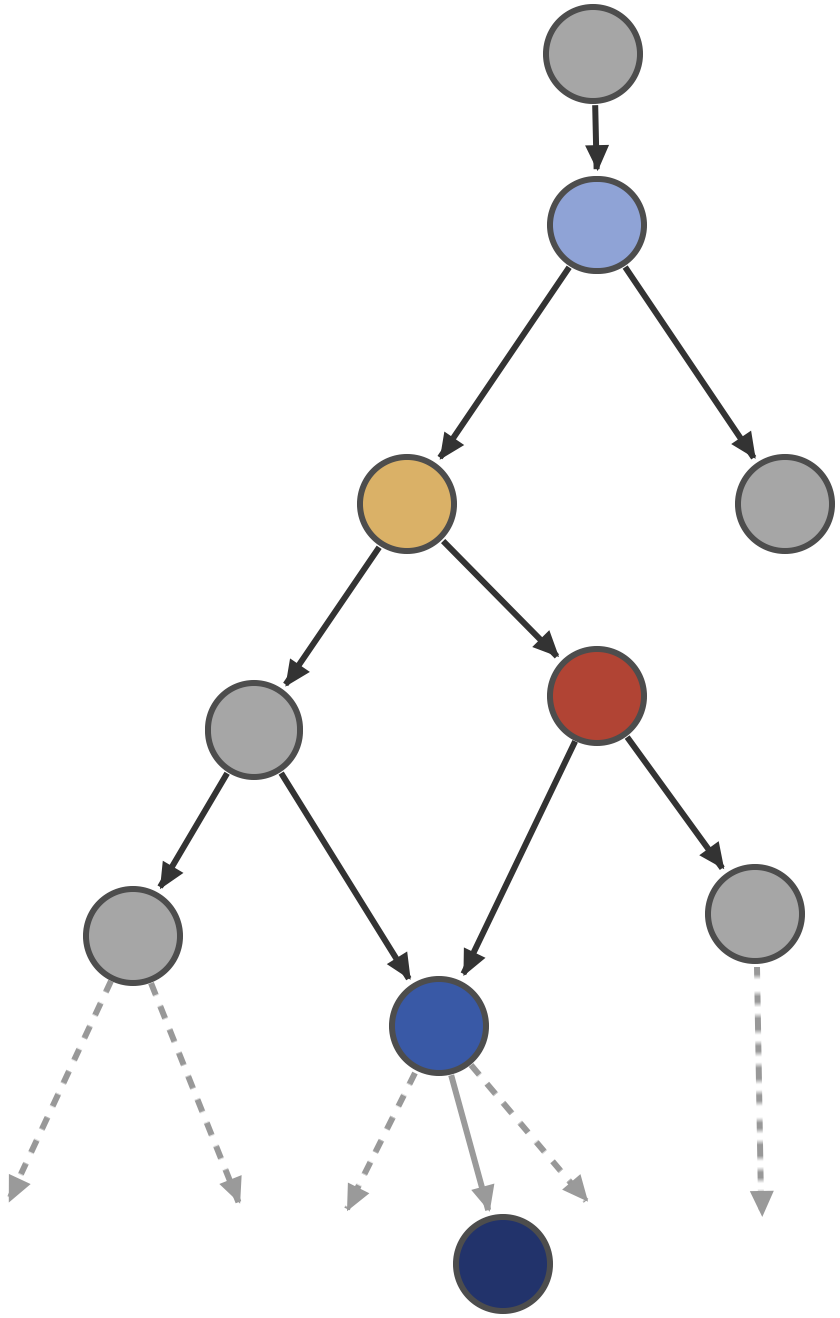
query	key	dot sim (x)	genuine? (y)
favor 	prepared 	$q_{\text{favor}}^{\top} k_{\text{prepare}}$	Yes (y=1)
favor 	water 	$q_{\text{favor}}^{\top} k_{\text{water}}$	No (y=0)

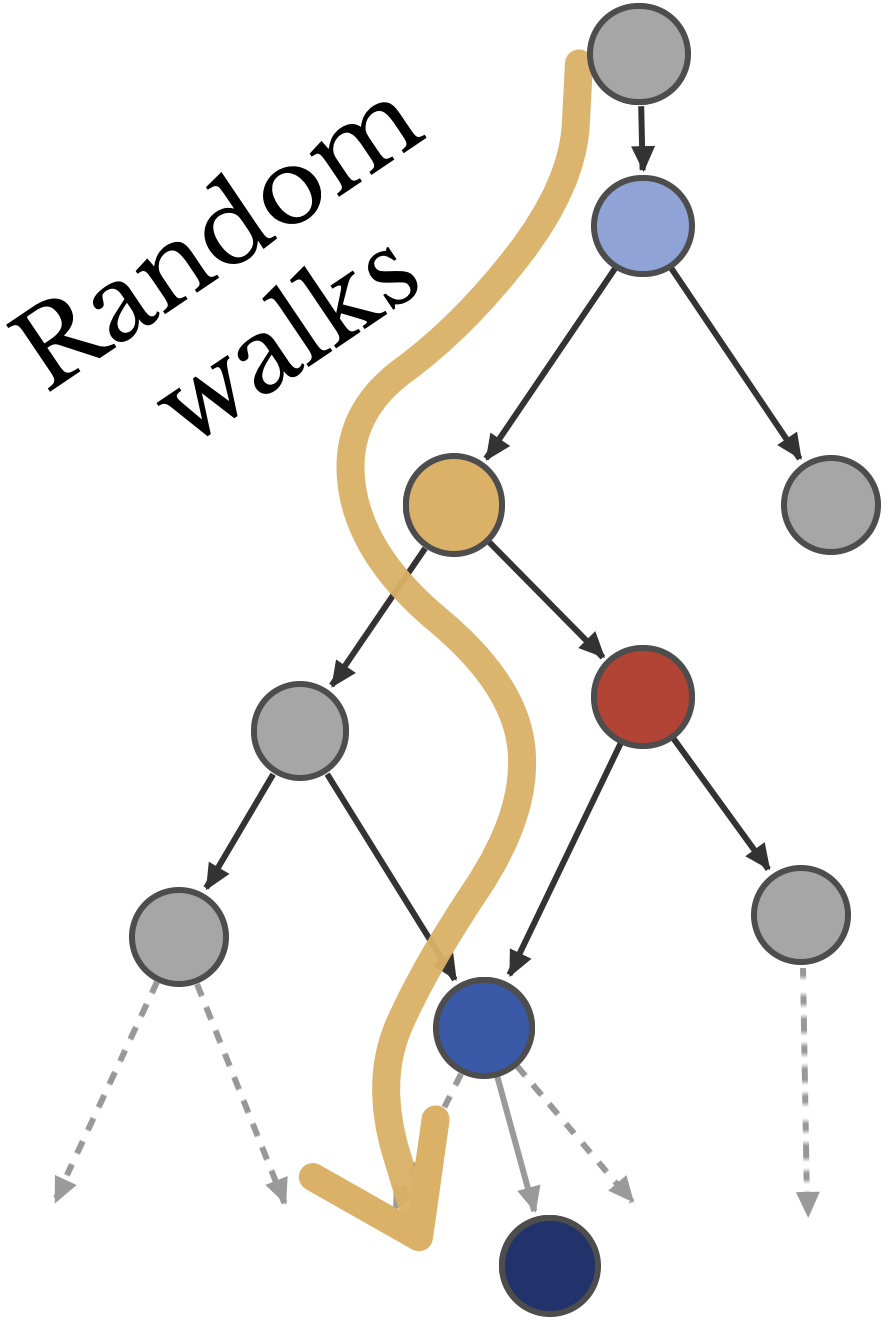
Classify by the logistic regression

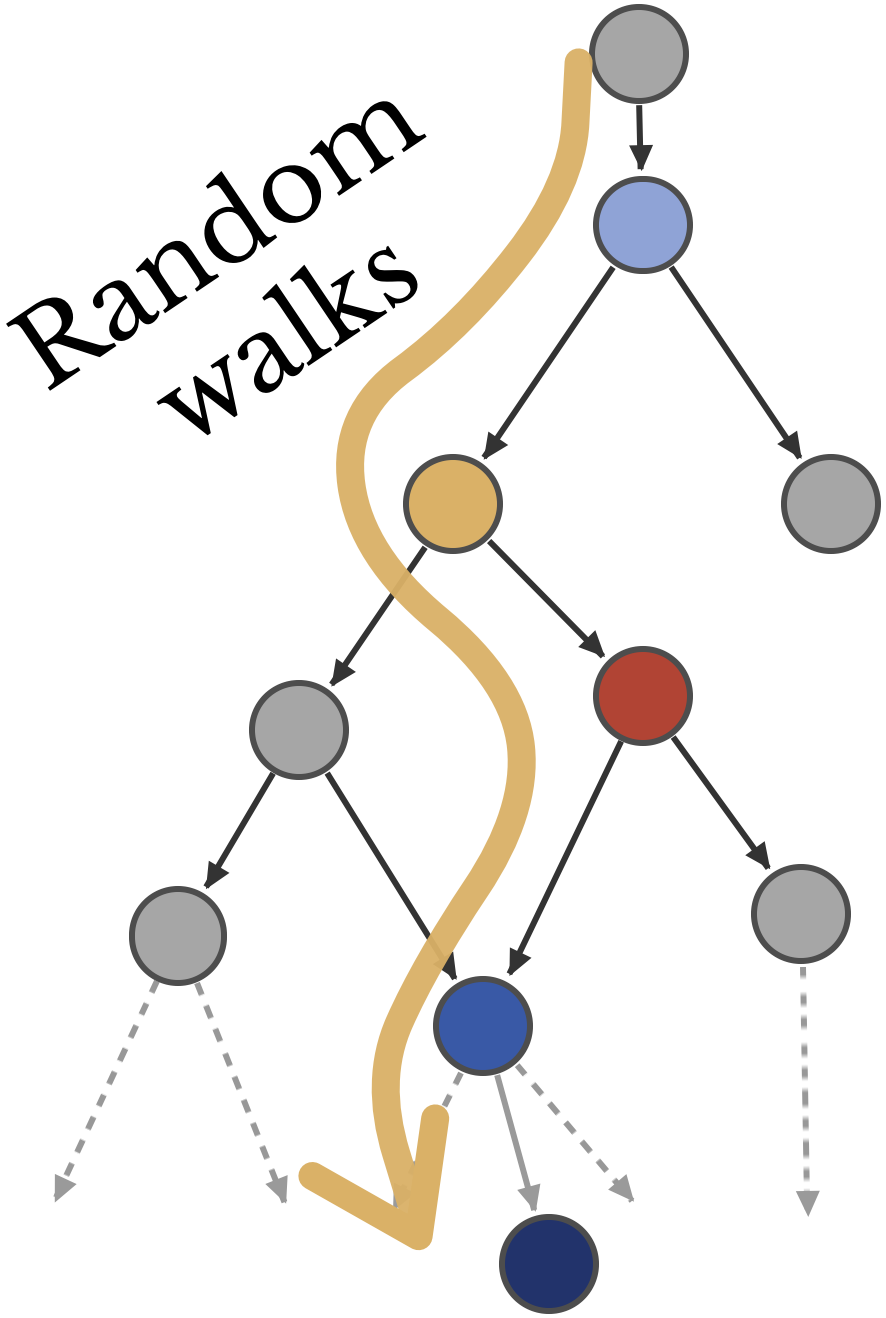
$$\sigma_\theta(i,j) = \frac{1}{1+\exp(-q_i^\top k_j)} \quad \theta = \{q_i, k_i\}_{i=1}^N$$

train by the log-likelihood maximization

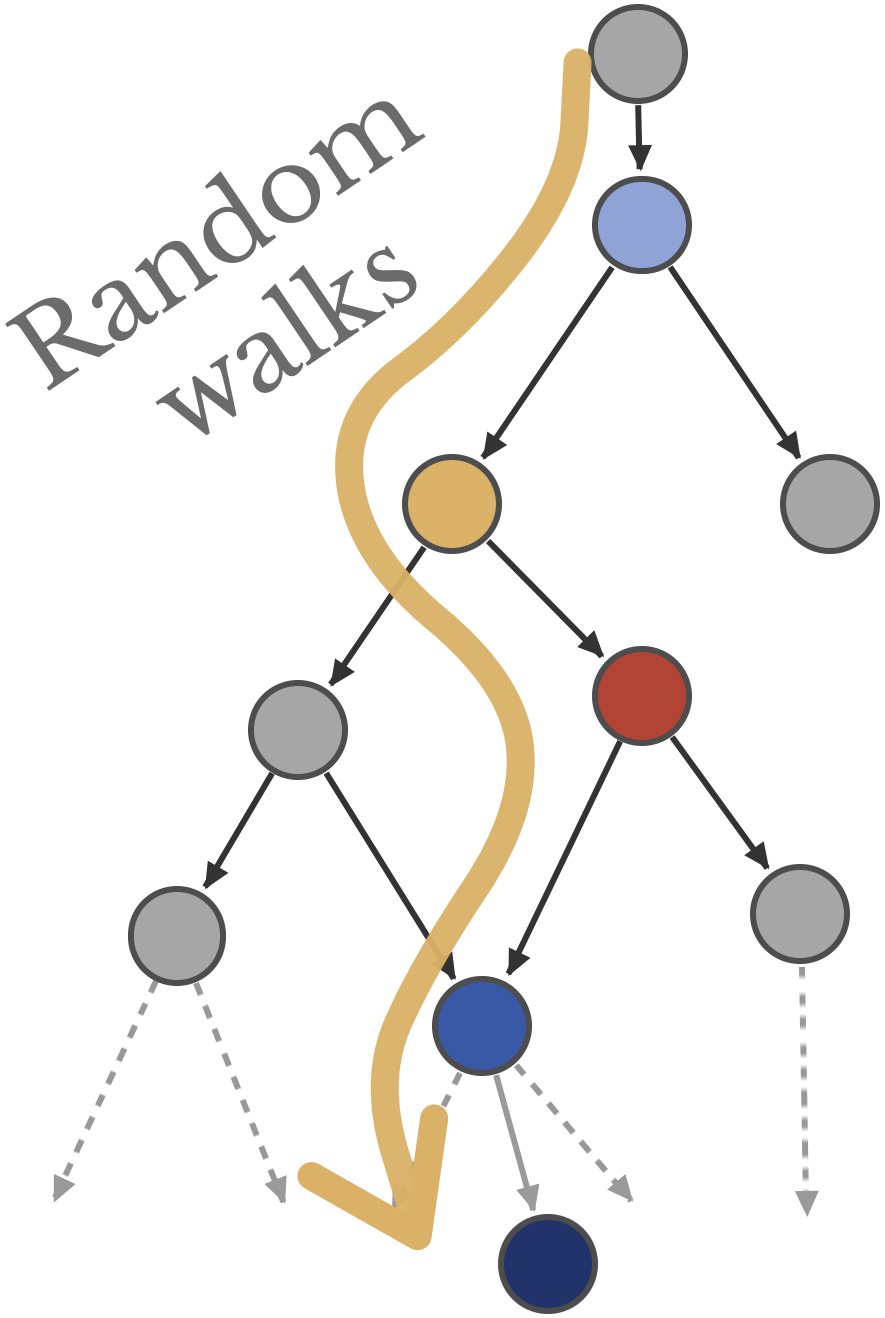
$$\max_\theta \sum_{(i,j)\in\mathcal{D}} [y_{i,j} \log \sigma_\theta(i,j) + (1-y_{i,j}) \log(1-\sigma_\theta(i,j))]$$

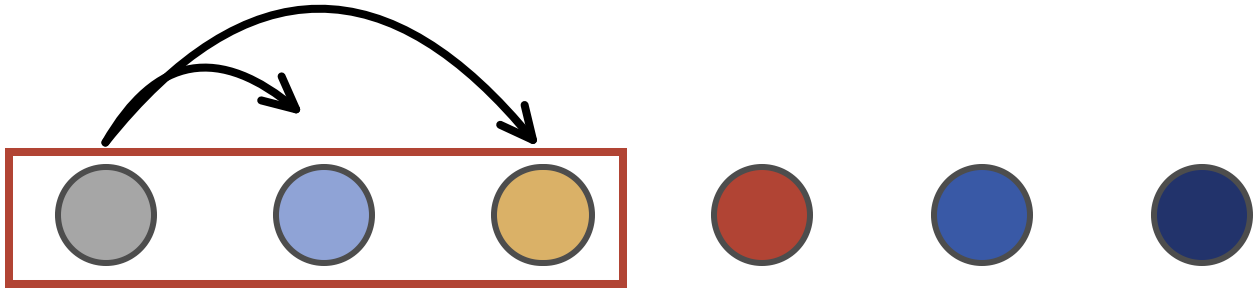



Random
walks


Random
walks

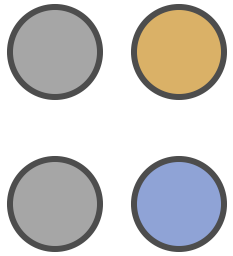




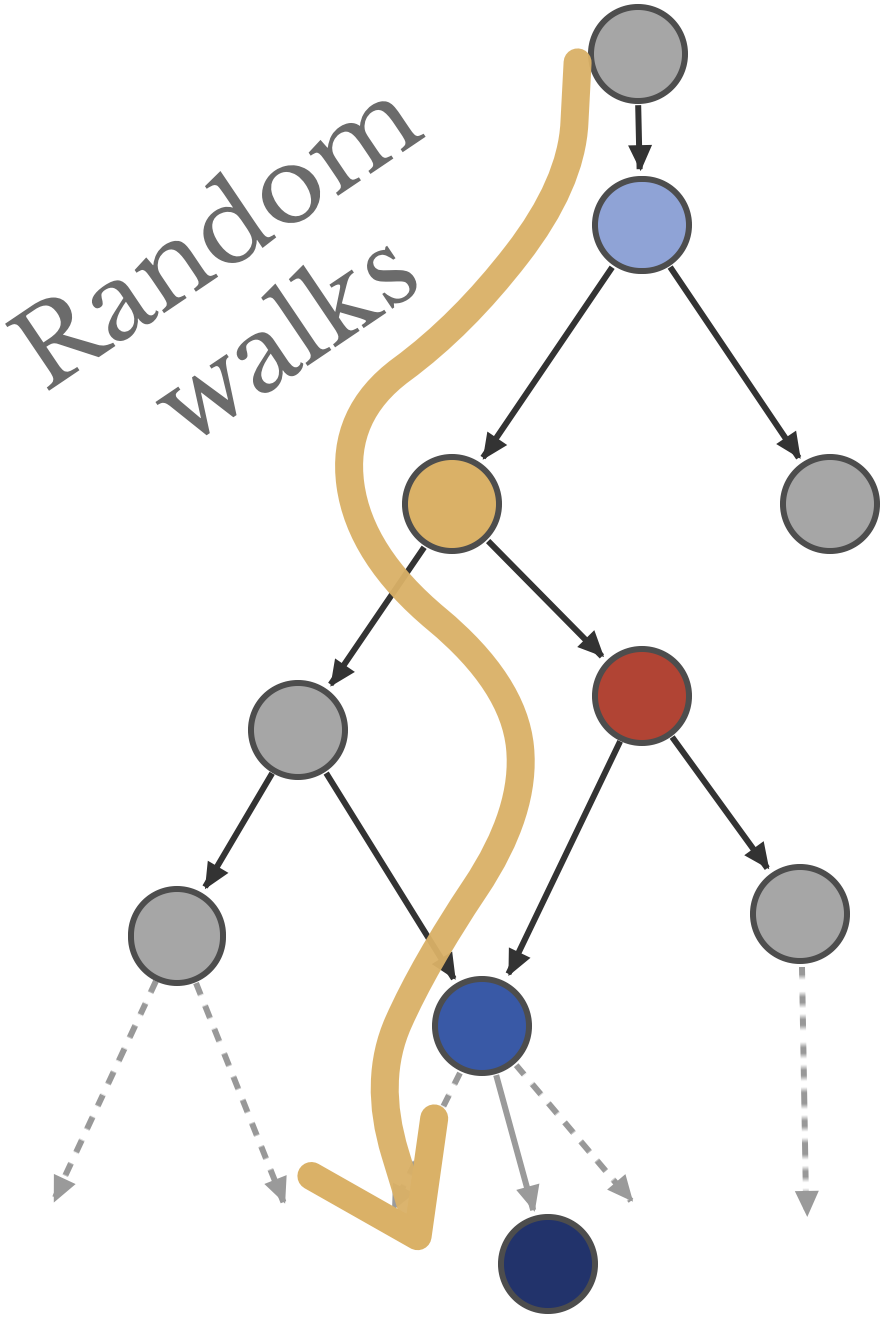


window

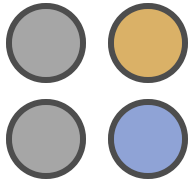
(query, key)

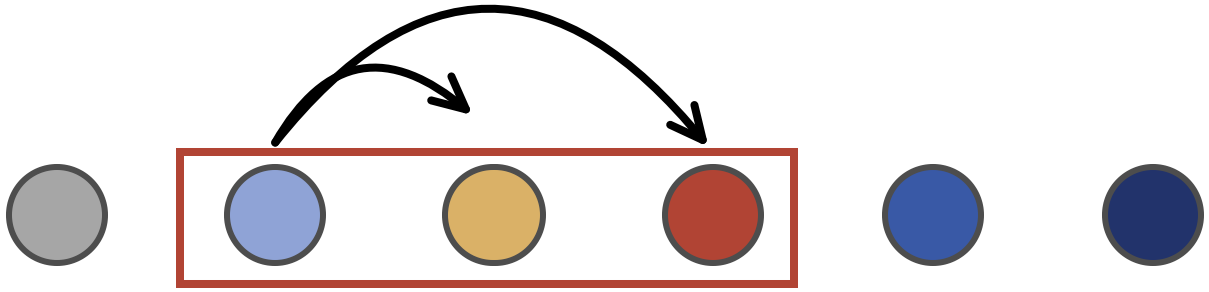


Training data

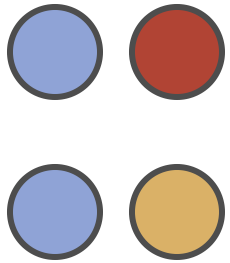


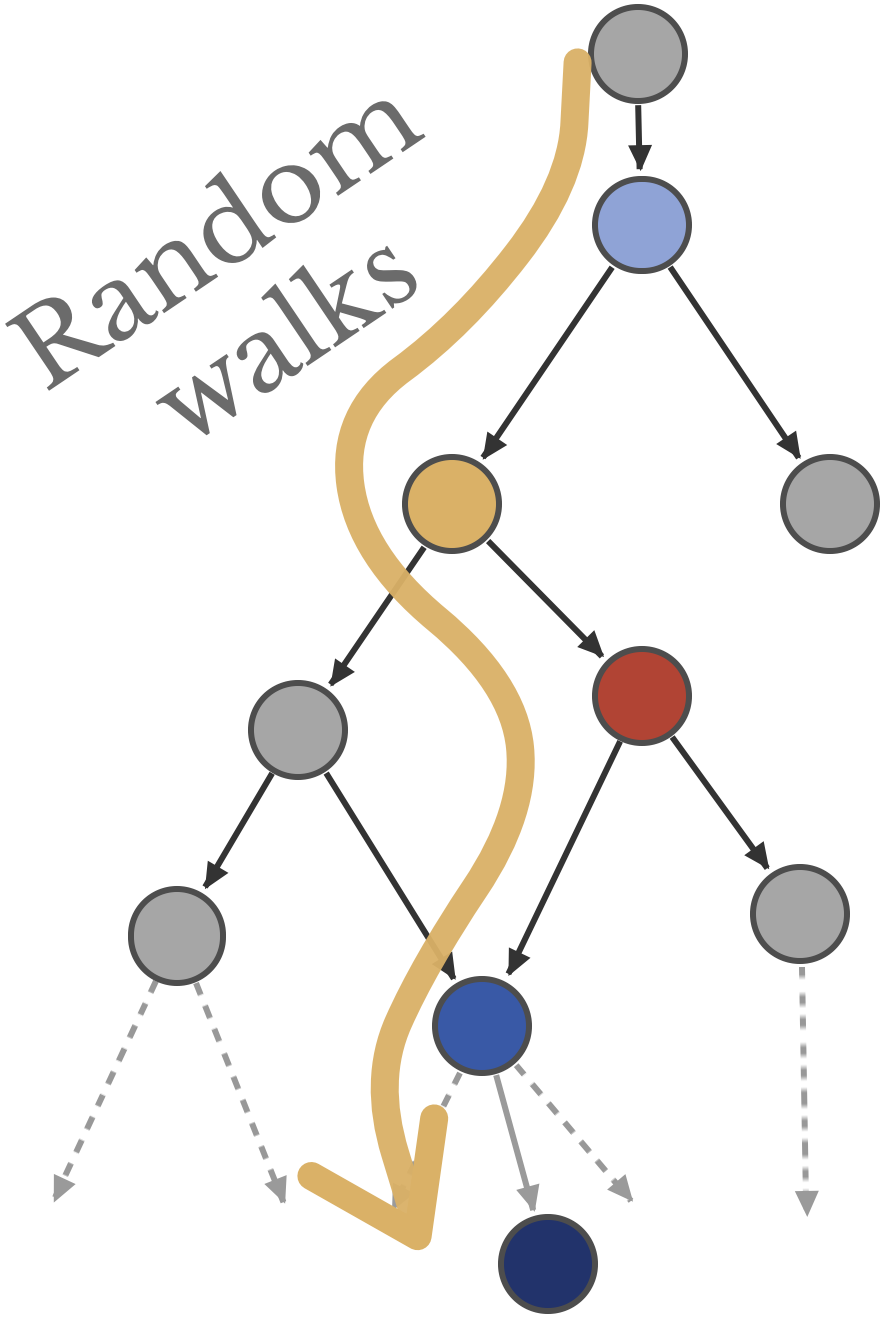
Training data

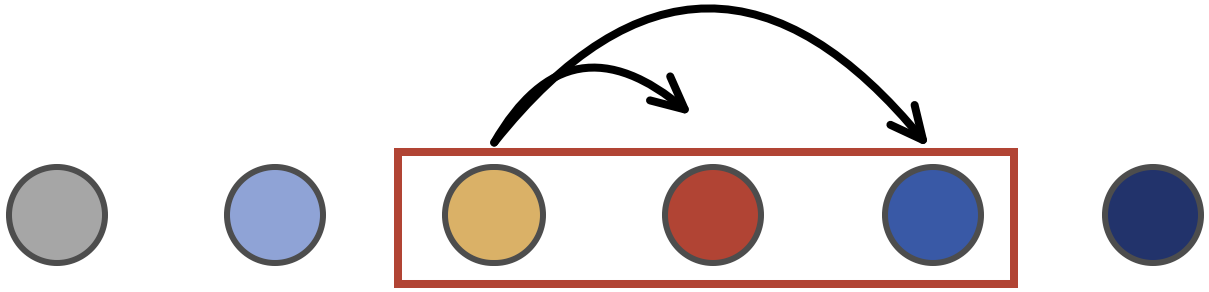




(query, key)



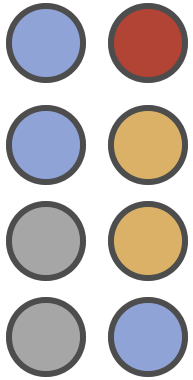


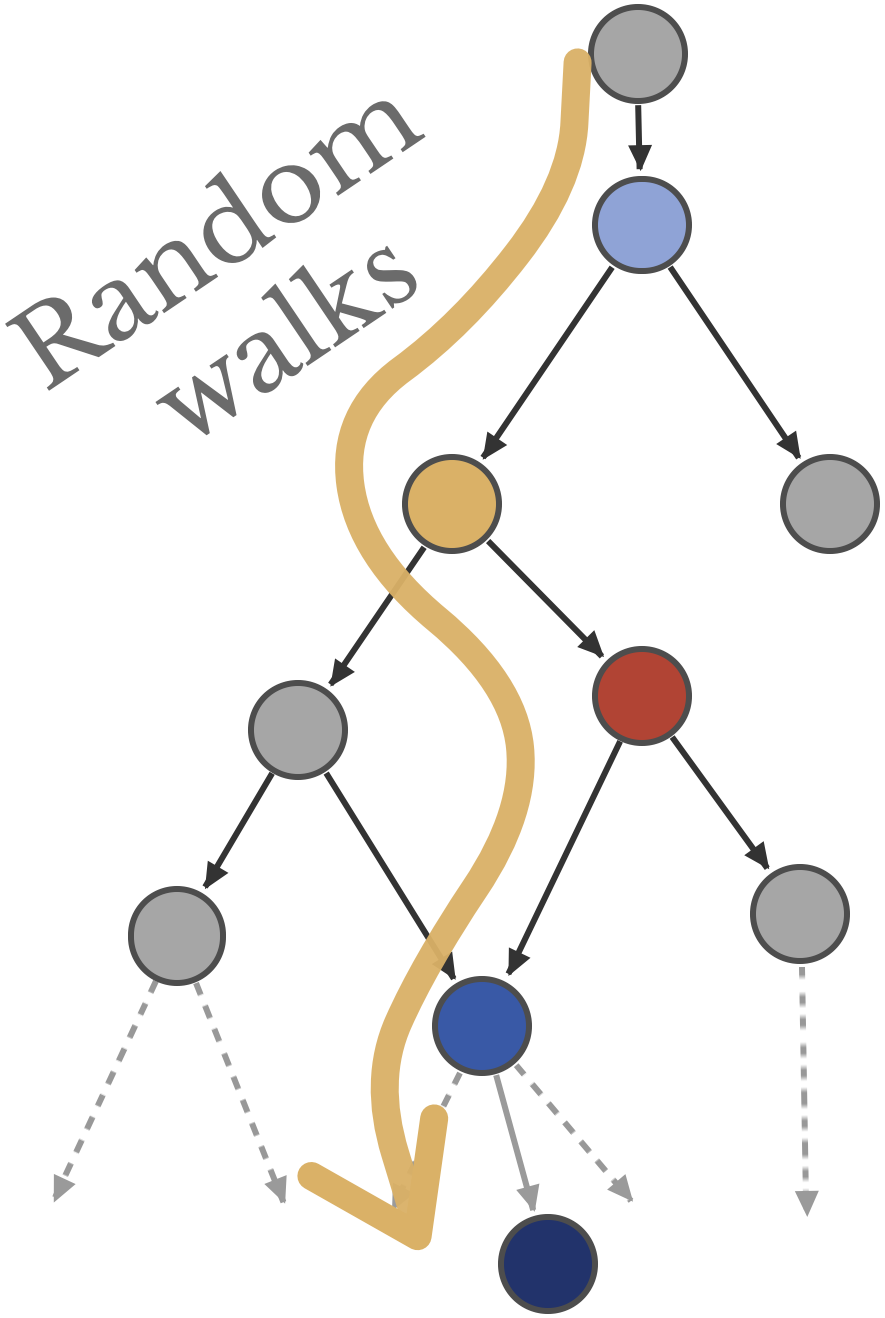


window

(query, key)

Training data



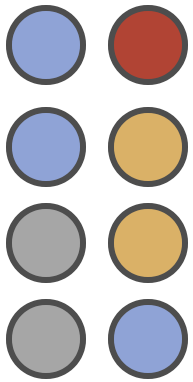


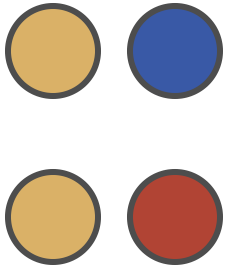


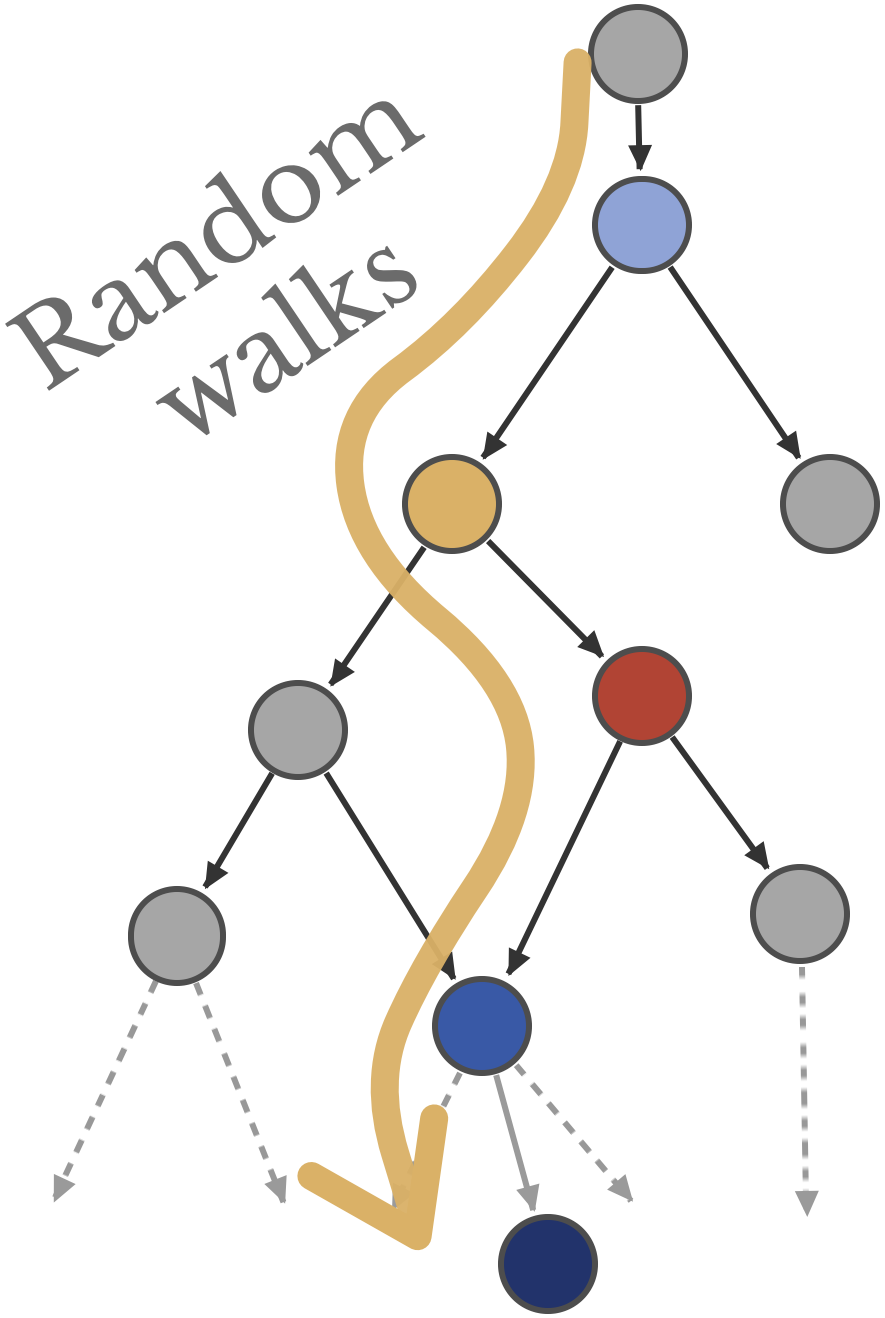
window

(query, key)

Training data



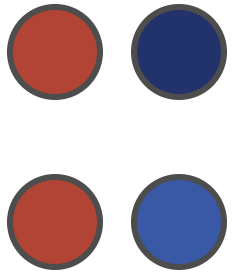


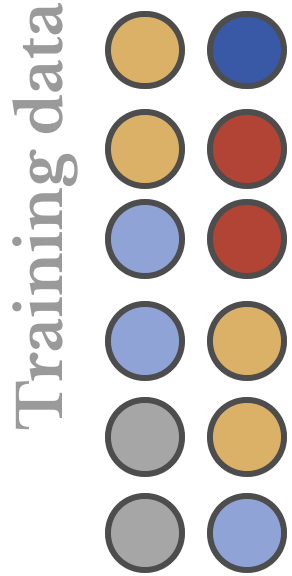




window

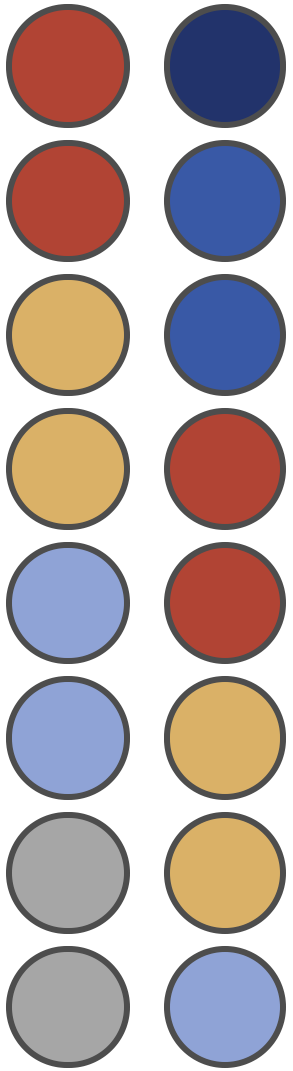
(query, key)



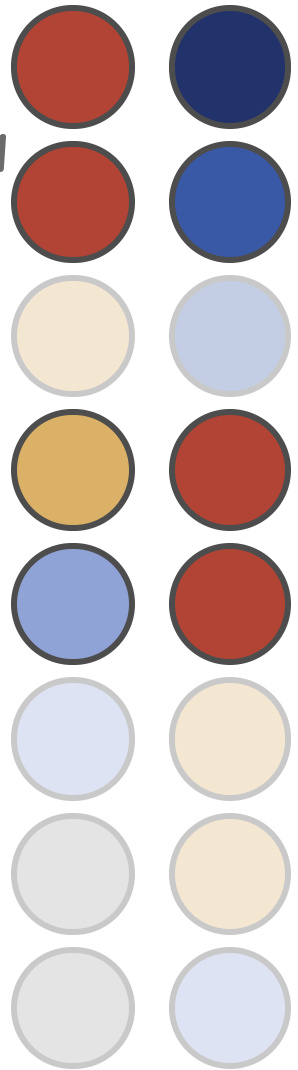


(query, key)

Training data

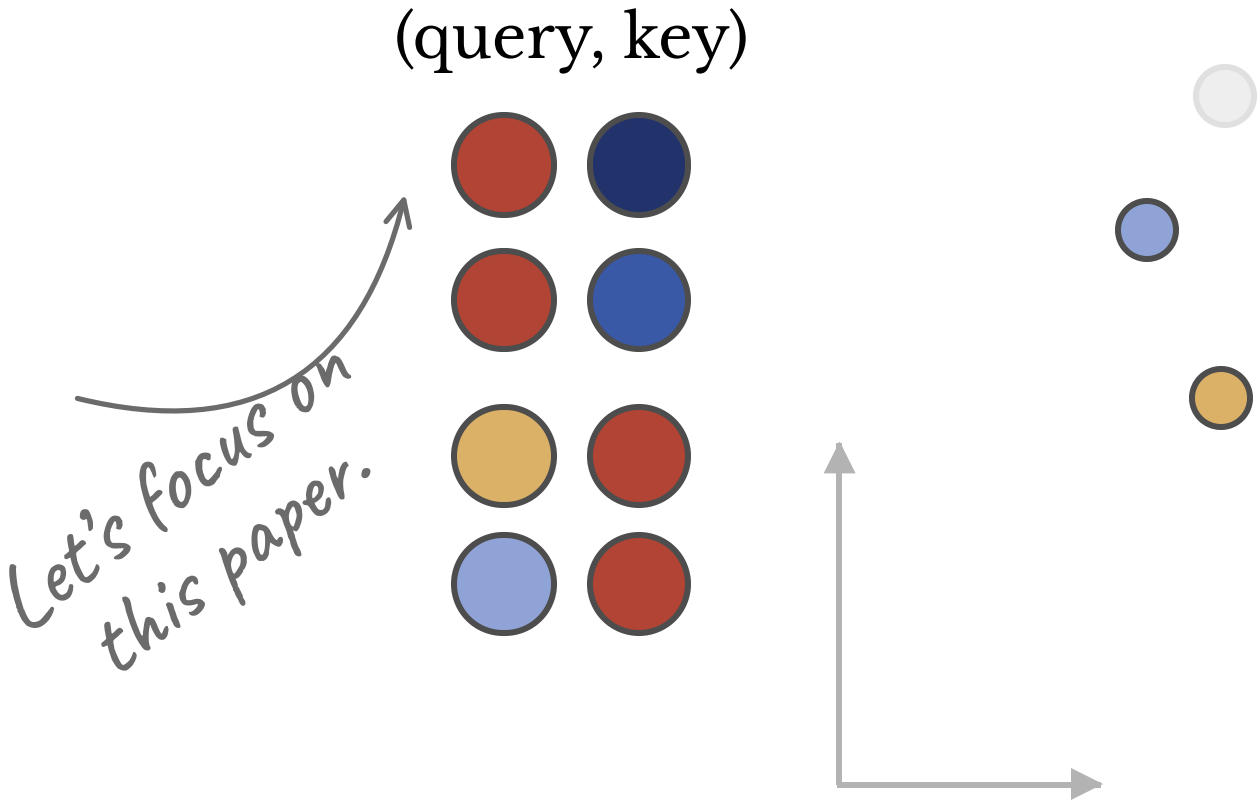


(query, key)



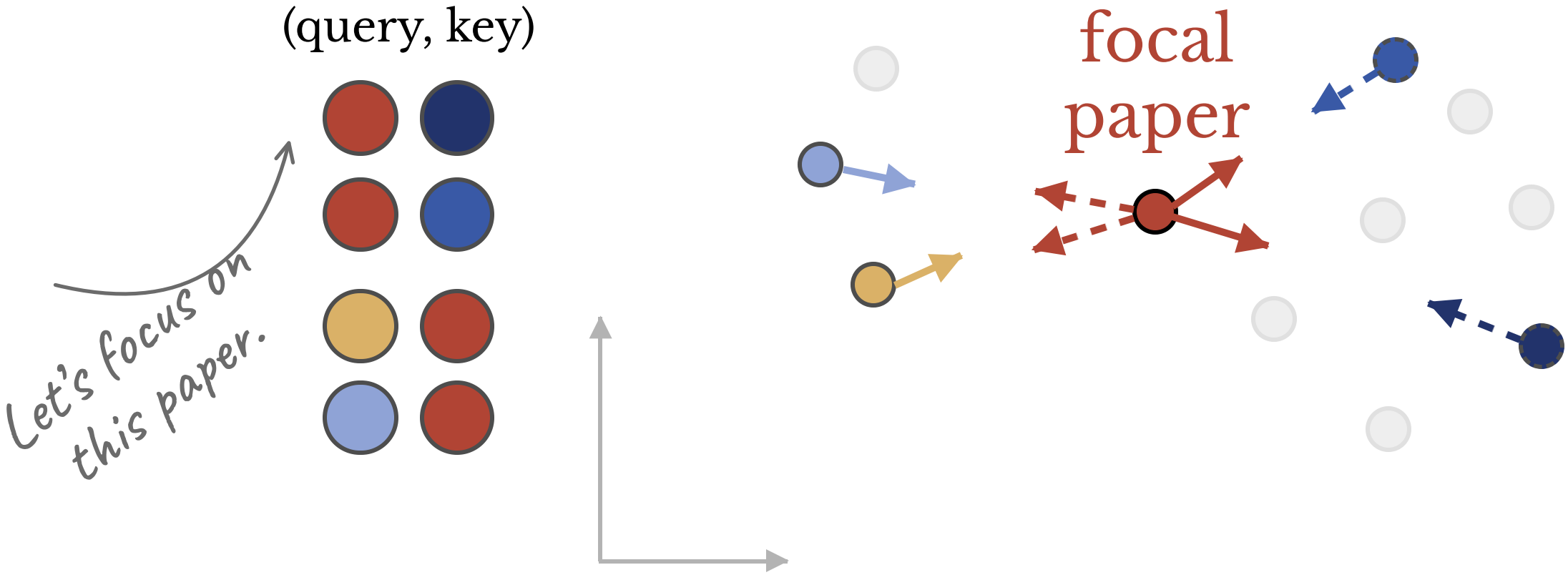
Let's focus on this paper.

irrelevant to the focal paper



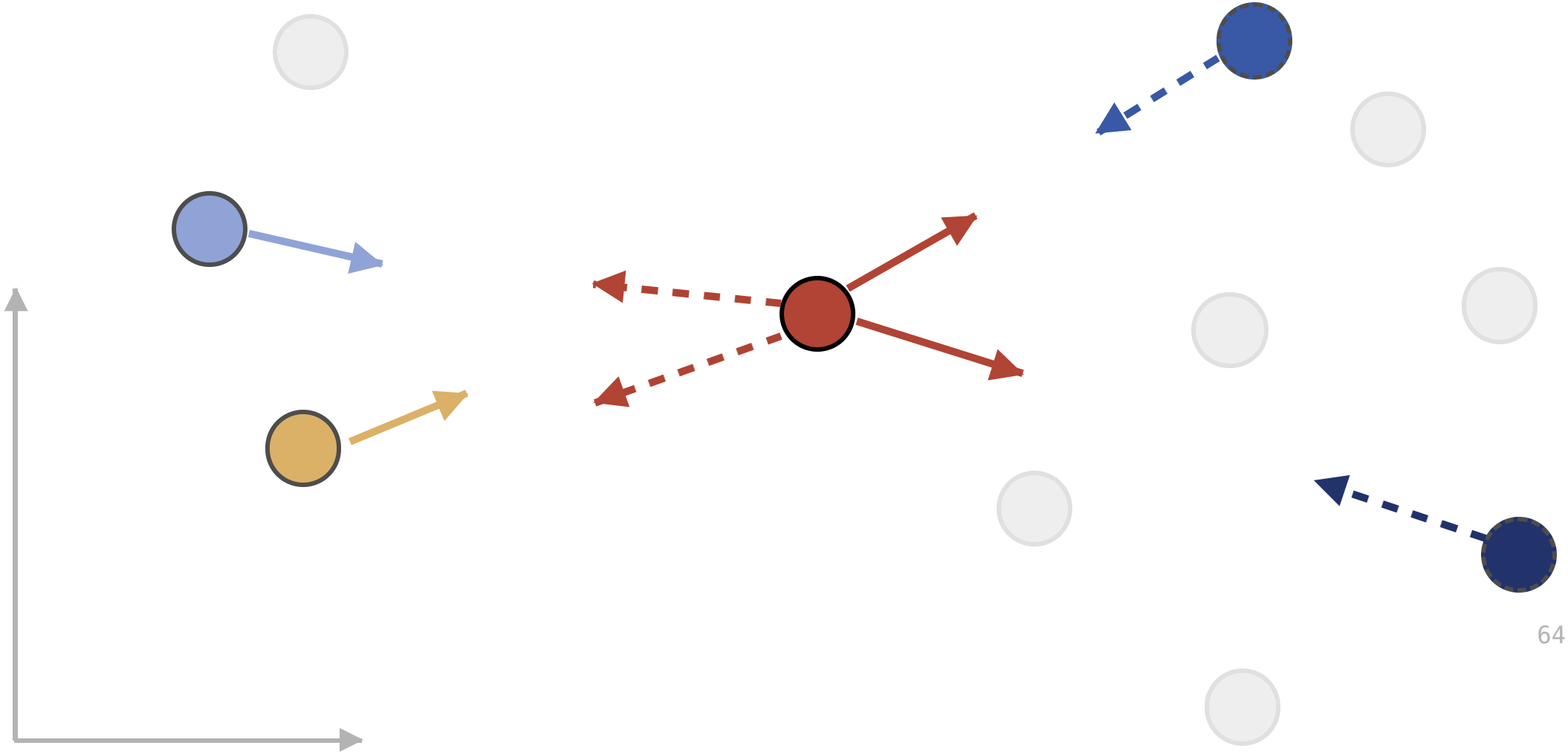


Paired query & key get closer.

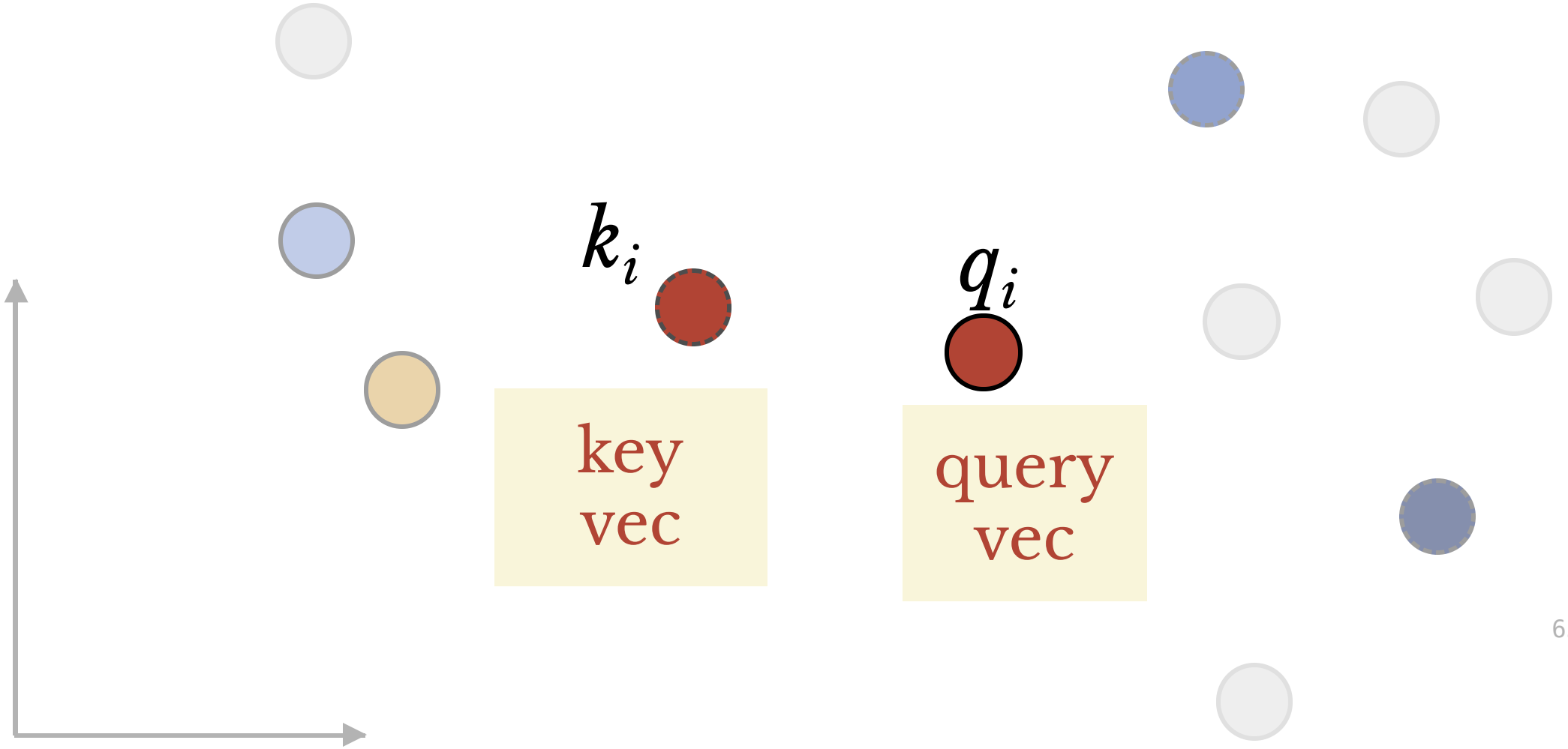


Paired query & key get closer.

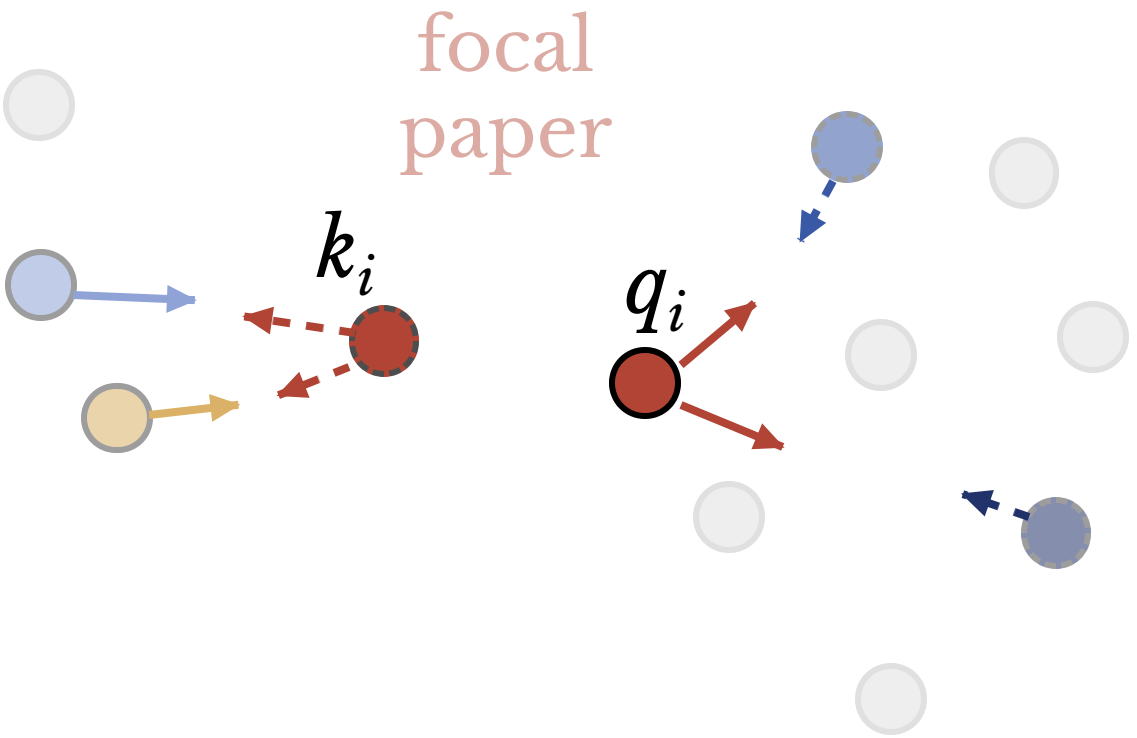
focal paper



focal
paper



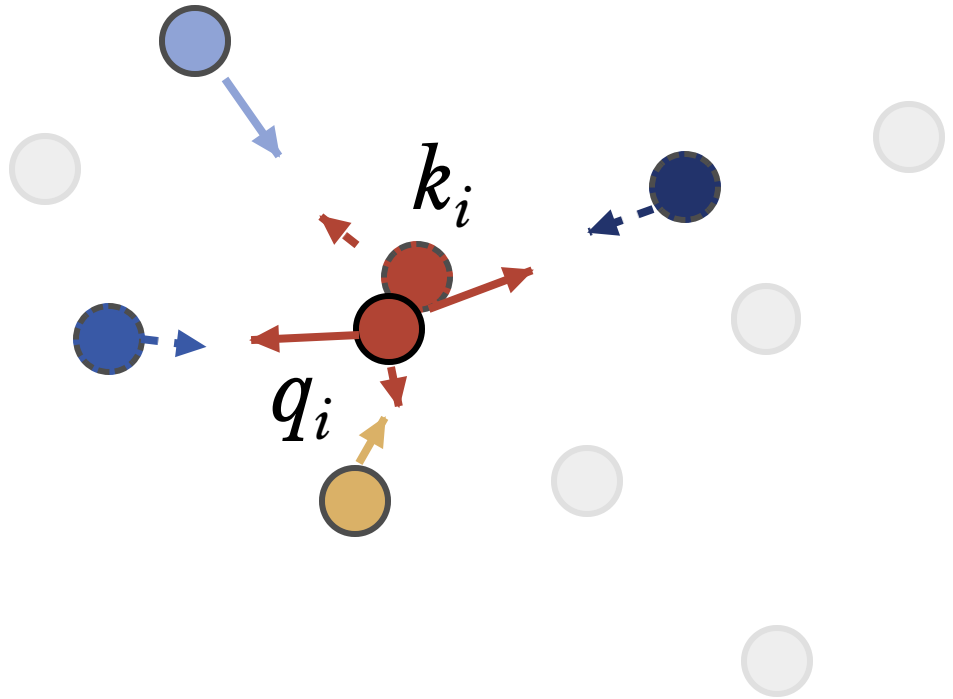
Not mixed



Large separation
of
key and query

~

Mixed



Discontinuity of citing
and cited papers

Embedding Disruptiveness Measure (EDM)

$$\Delta_i := 1 - \cos(q_i, k_i)$$

Dissimilarity of the descendants and ascendants of node i.

Δ = 1.002
Top 4.3 %



Δ = 1.005
Top 4.1 %

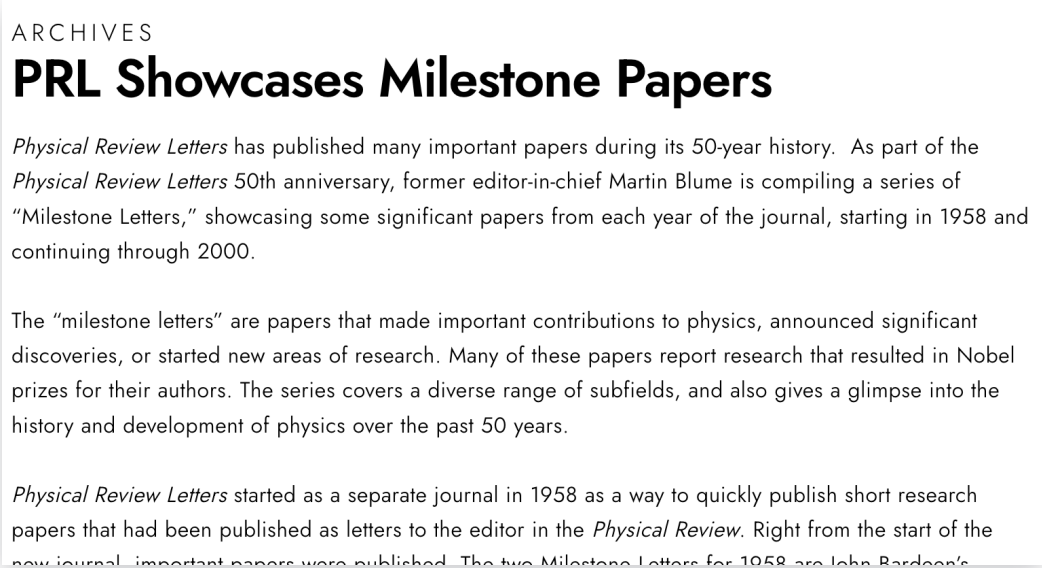
Does EDM beat the D-index? 🤔

Tested with APS data


APSNews

May 2008 (Volume 17, Number 5)

PRL Showcases Milestone Papers


ARCHIVES

PRL Showcases Milestone Papers

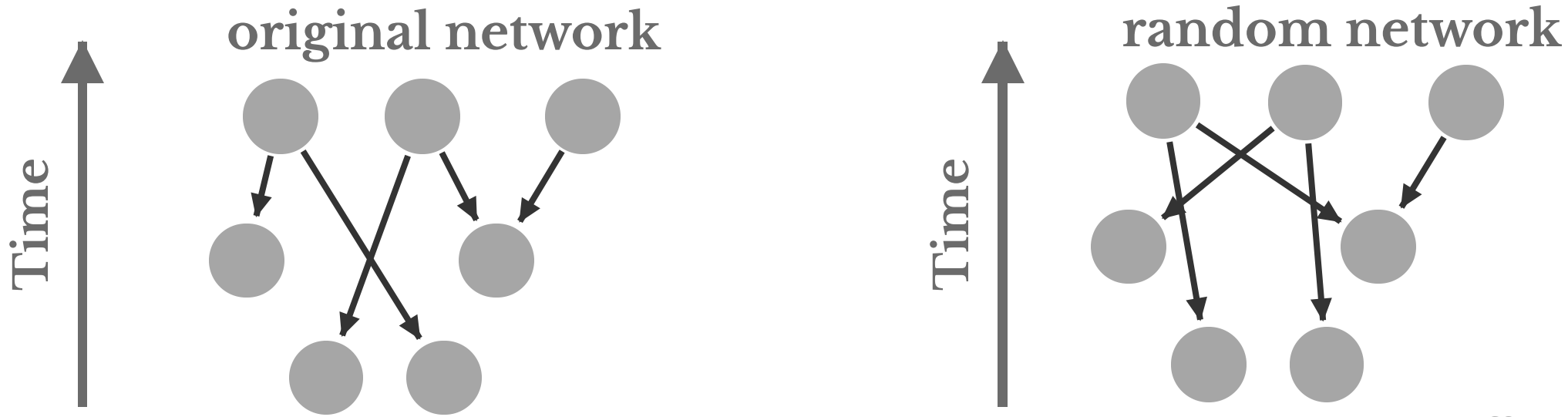
Physical Review Letters has published many important papers during its 50-year history. As part of the *Physical Review Letters* 50th anniversary, former editor-in-chief Martin Blume is compiling a series of "Milestone Letters," showcasing some significant papers from each year of the journal, starting in 1958 and continuing through 2000.

The "milestone letters" are papers that made important contributions to physics, announced significant discoveries, or started new areas of research. Many of these papers report research that resulted in Nobel prizes for their authors. The series covers a diverse range of subfields, and also gives a glimpse into the history and development of physics over the past 50 years.

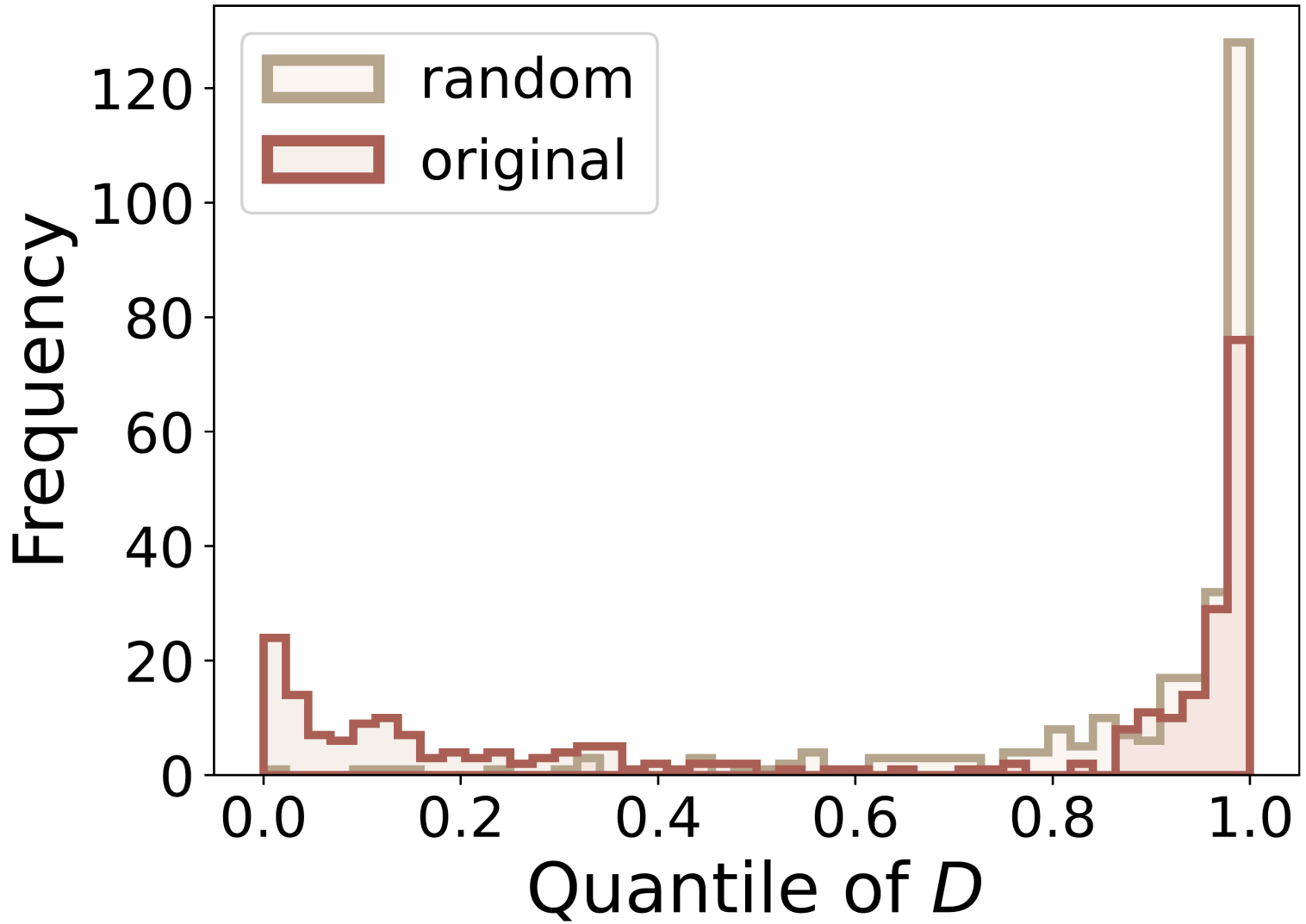
Physical Review Letters started as a separate journal in 1958 as a way to quickly publish short research papers that had been published as letters to the editor in the *Physical Review*. Right from the start of the new journal, important papers were published. The two Milestone Letters for 1958 are John Bardeen's

Conformal analysis

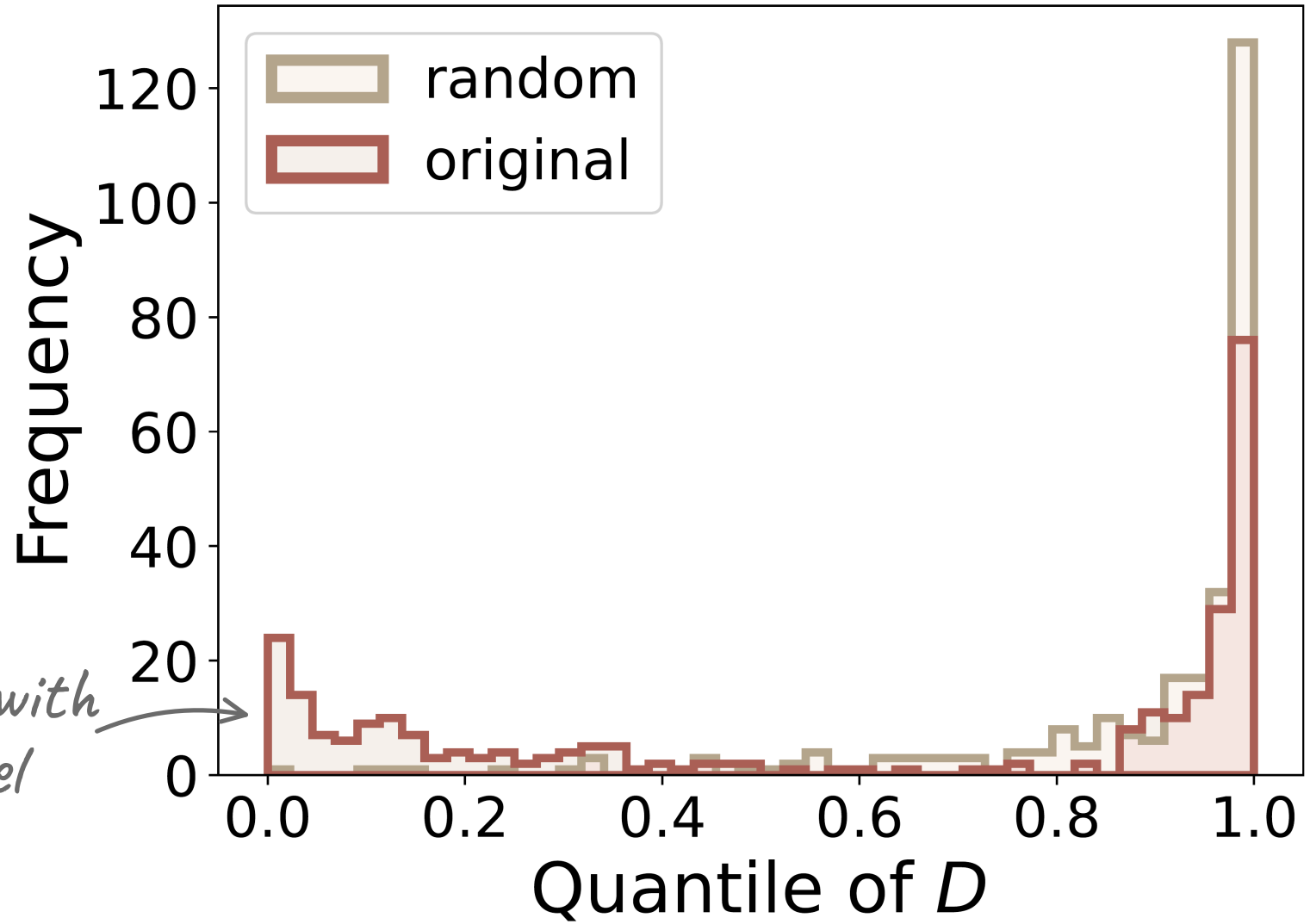
Preserve degrees and time gap.
Shuffle everything else.



D-index of milestone works (APS)

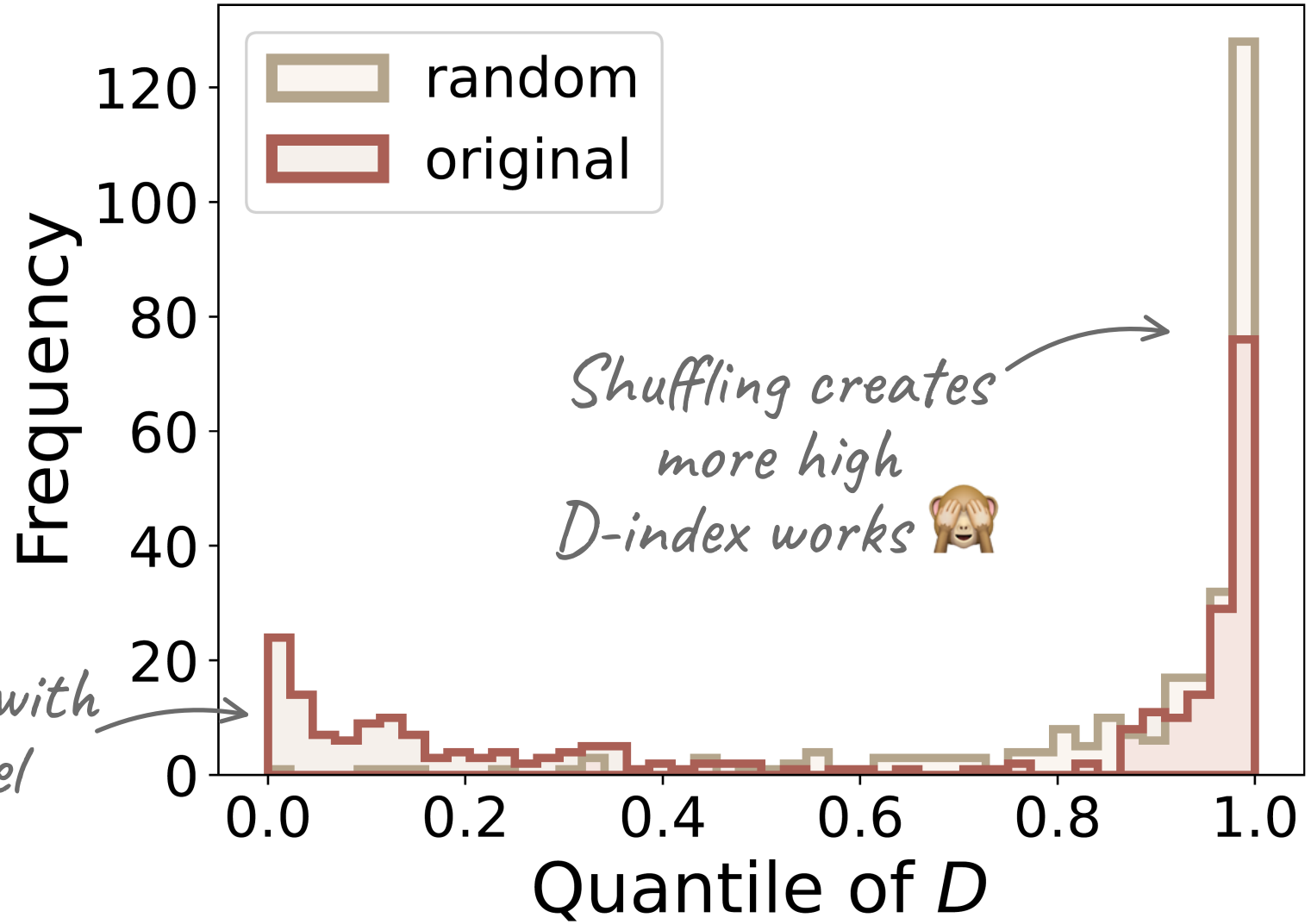


D-index of milestone works (APS)



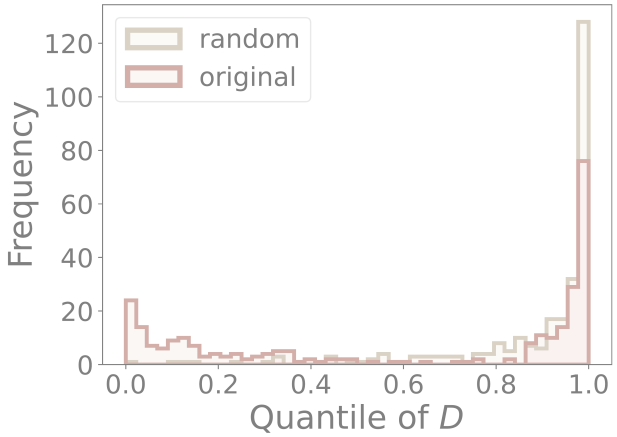
Milestone papers with the bottom-level D-index value.

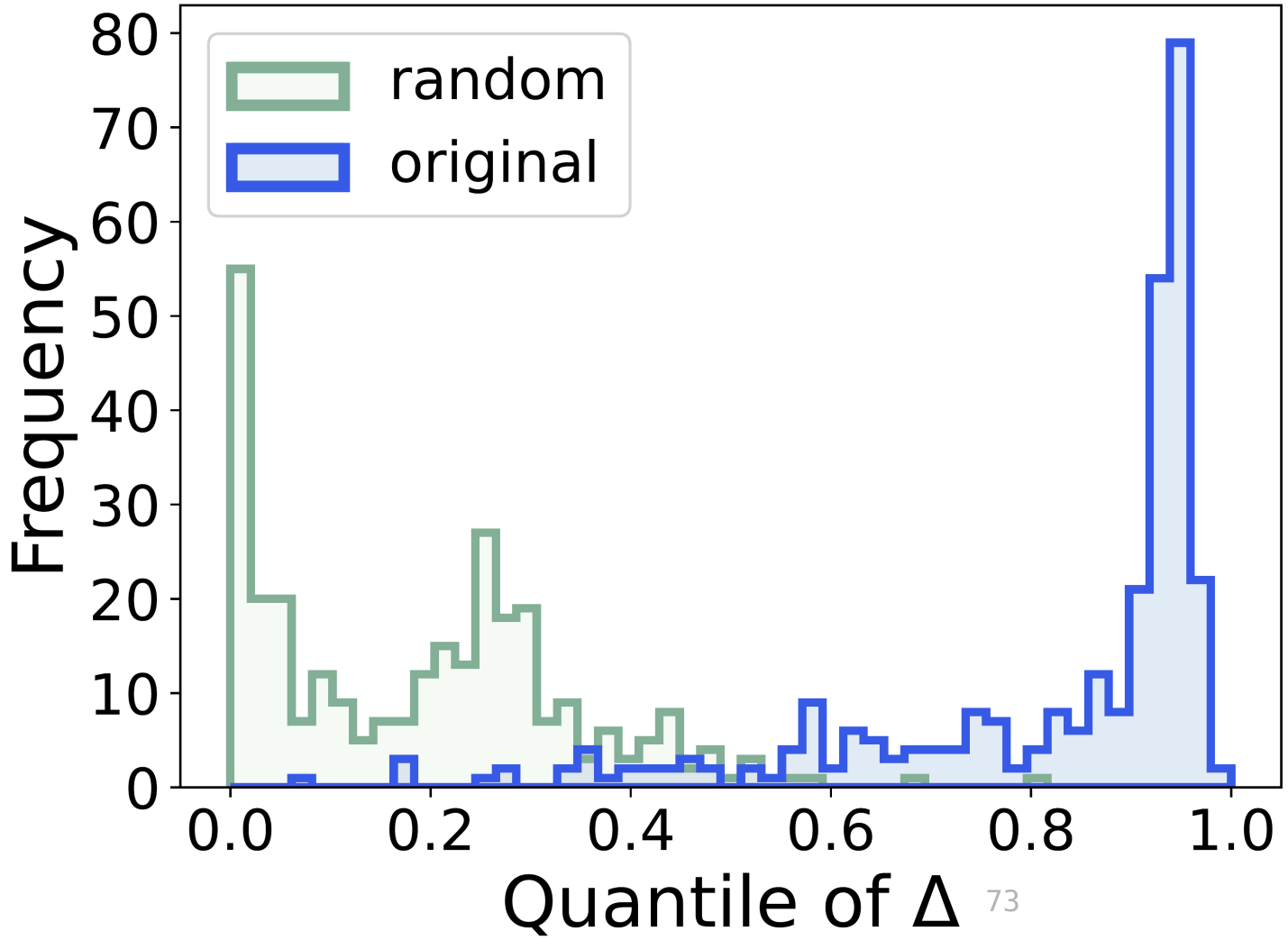
D-index of milestone works (APS)



EDM

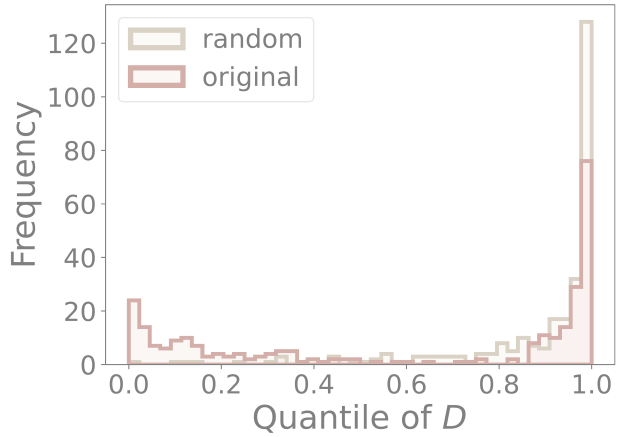
D-index

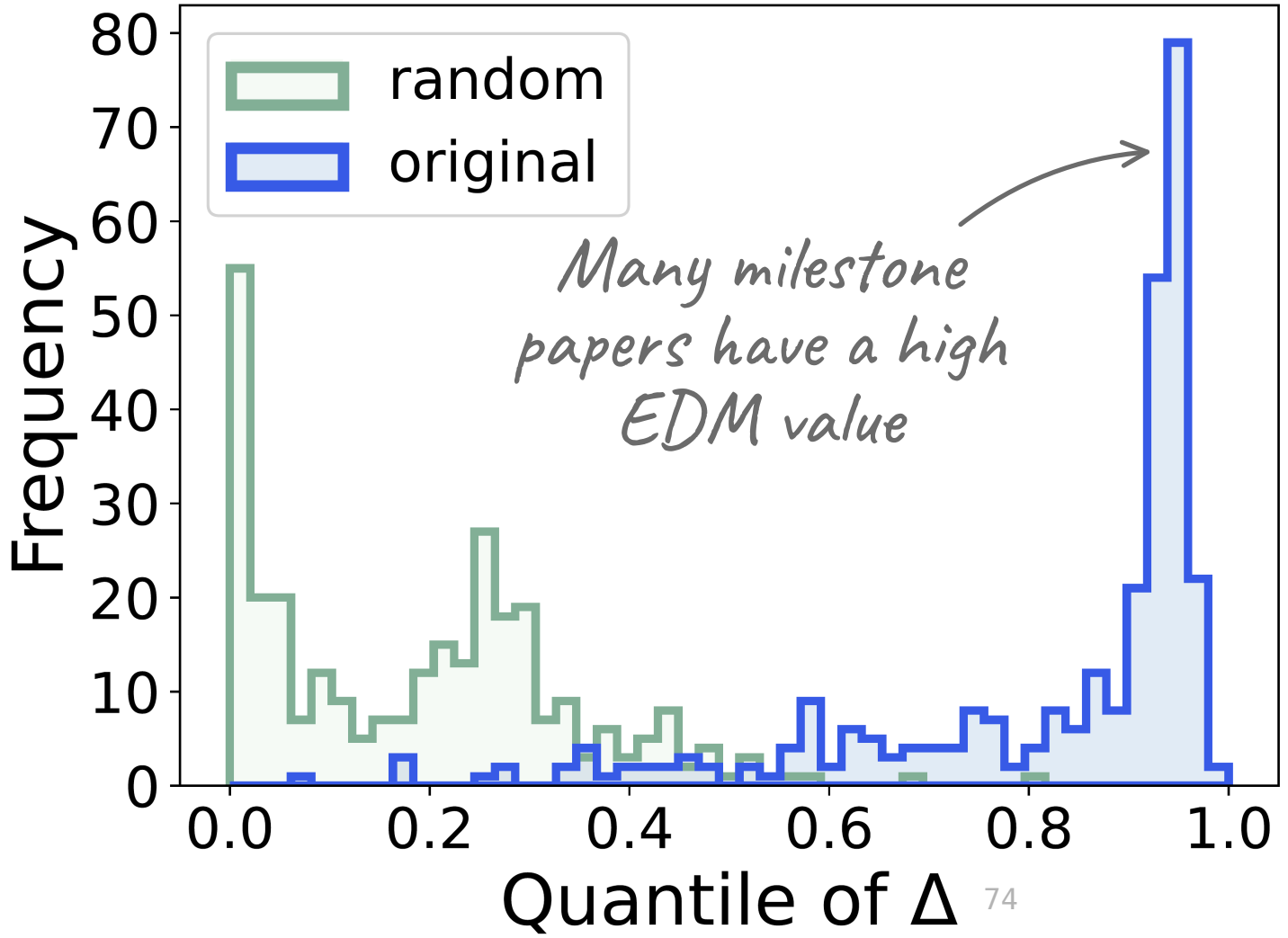

random
original
Frequency
120
100
80
60
40
20
0
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0.2
0.4
0.6
0.8
1.0
Quantile of D


random
original
Frequency
80
70
60
50
40
30
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10
0
0.0
0.2
0.4
0.6
0.8
1.0
Quantile of Δ

EDM

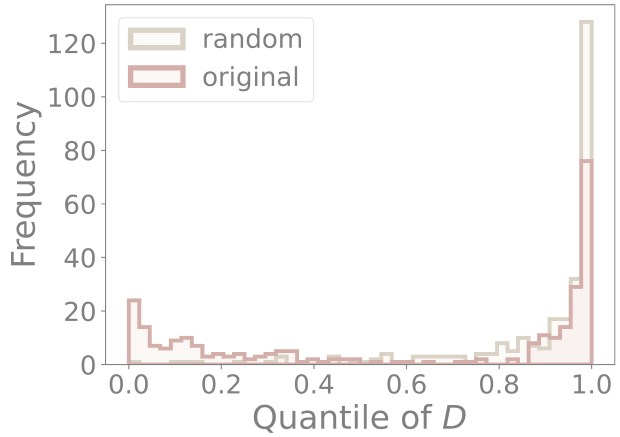
D-index

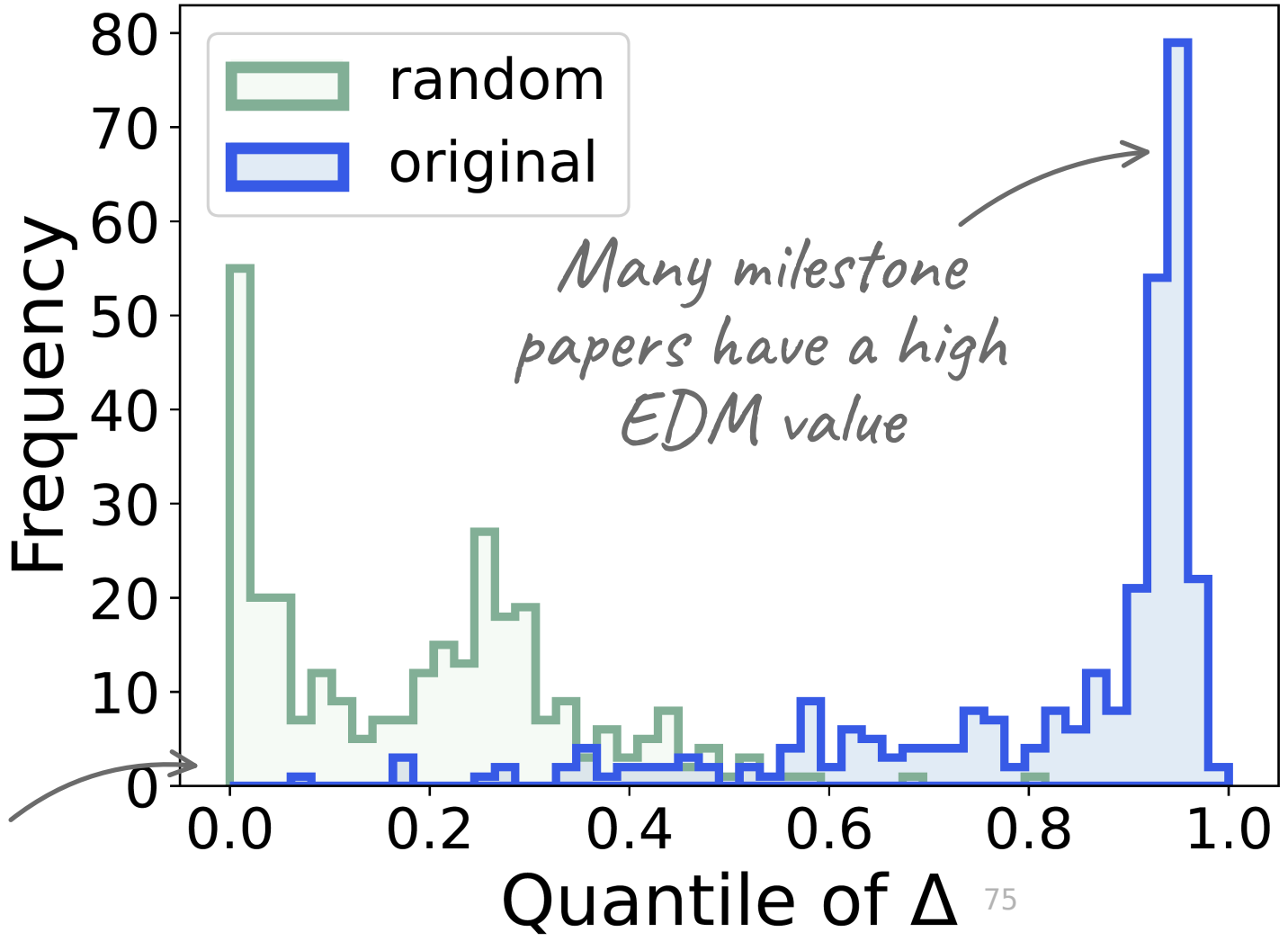

random
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Frequency
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Quantile of D


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original
Frequency
80
70
60
50
40
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Quantile of Δ
Many milestone papers have a high EDM value

EDM

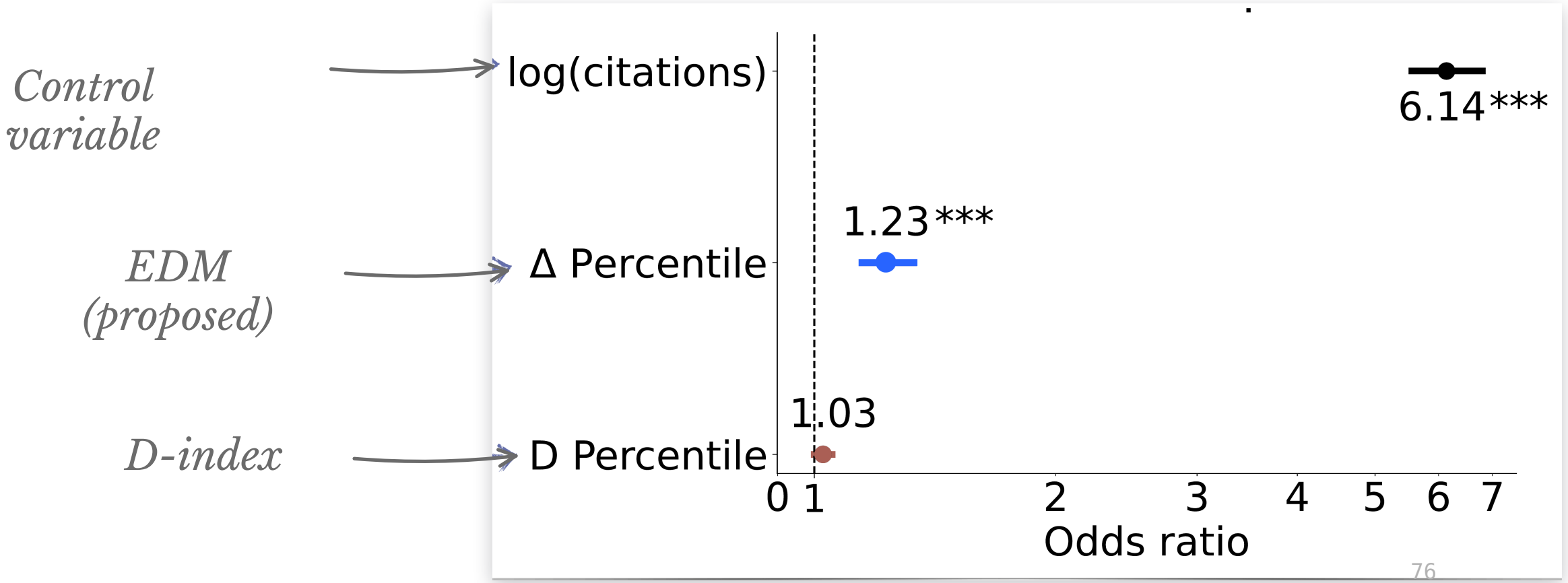
D-index





Randomization reduces the EDM-score for most works

EDM aligns better with human judgment



APSNews

May 2008 (Volume 17, Number 5)

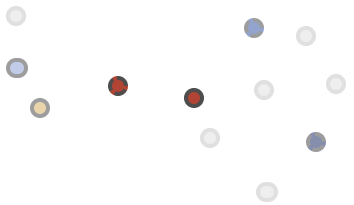
PRL Showcases Milestone Papers

Summary



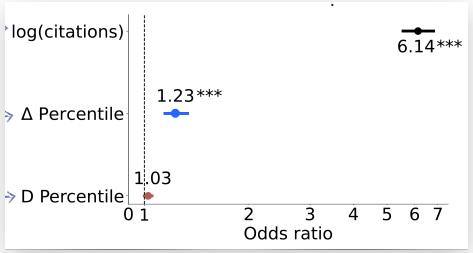
- **The D-index fails on milestone works**

  Higgs (1964): D = −0.27, Bottom 0.1 %



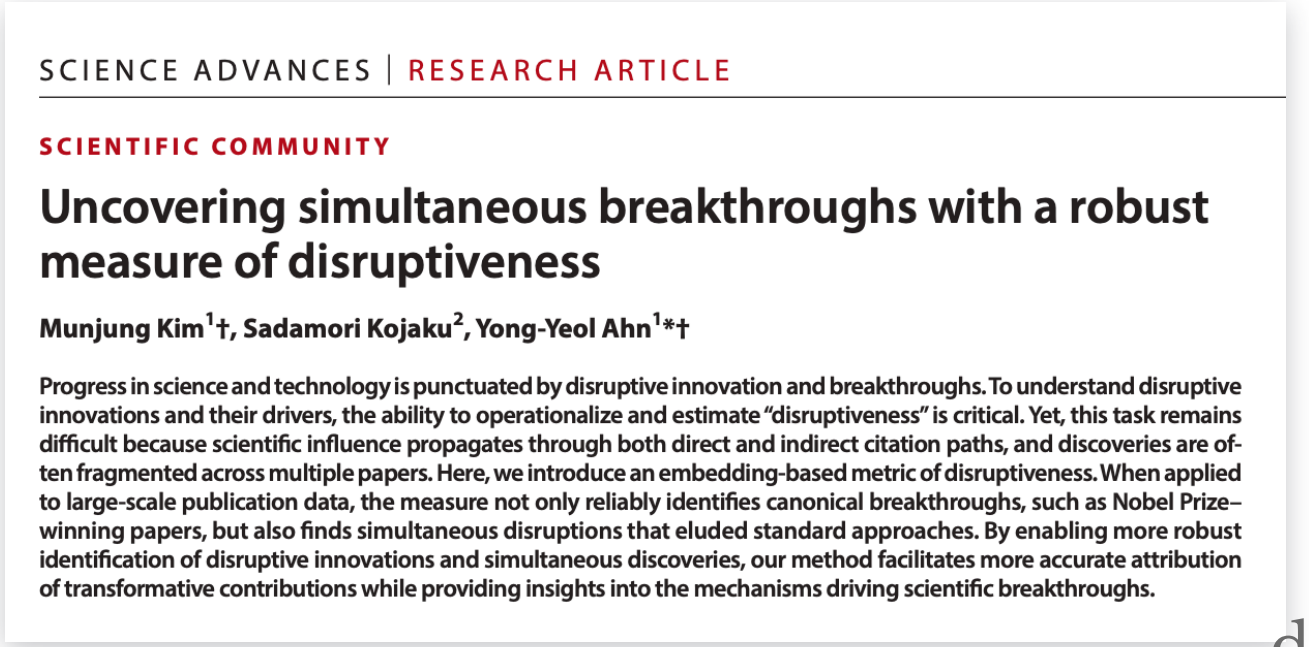
- **EDM preserves continuity**

  A large query-key gap marks a discontinuity



- **EDM matches human judgment**

  Odds ratio 1.23, against 1.03 for the D-index



SCIENCE ADVANCES | RESEARCH ARTICLE

SCIENTIFIC COMMUNITY

Uncovering simultaneous breakthroughs with a robust measure of disruptiveness

Munjung Kim[1]†, Sadamori Kojaku[2], Yong-Yeol Ahn[1]*†

Progress in science and technology is punctuated by disruptive innovation and breakthroughs. To understand disruptive innovations and their drivers, the ability to operationalize and estimate "disruptiveness" is critical. Yet, this task remains difficult because scientific influence propagates through both direct and indirect citation paths, and discoveries are often fragmented across multiple papers. Here, we introduce an embedding-based metric of disruptiveness. When applied to large-scale publication data, the measure not only reliably identifies canonical breakthroughs, such as Nobel Prize–winning papers, but also finds simultaneous disruptions that eluded standard approaches. By enabling more robust identification of disruptive innovations and simultaneous discoveries, our method facilitates more accurate attribution of transformative contributions while providing insights into the mechanisms driving scientific breakthroughs.

Read the paper

doi.org/10.1126/sciadv.adx3420

Science Advances 12, eadx3420 (2026)



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